United States Patent Application Publication	20250286663
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	Esswie; Ali

ADAPTIVE NON-TERRESTRIAL PACKET RETRANSMISSIONS

Abstract

A user equipment may receive downlink traffic packets according to a communication session with a non-terrestrial network node. The user equipment may request that retransmission of payload packet(s) transmitted by the non-terrestrial node to the user equipment that is/are unsuccessfully decoded by the user equipment be delegated to a terrestrial radio access network node. The user equipment may include modulation and coding scheme information and redundancy version information in a request to the terrestrial node for delegated retransmission of unsuccessfully decoded packet received from the non-terrestrial node. The terrestrial node may request and receive, from core network equipment or non-terrestrial node equipment, payload packet(s) for which the user equipment requested delegated retransmission, and may retransmit, to the user equipment, the requested payload according to the modulation and coding scheme and a next redundancy value with respect to transmission by the non-terrestrial node of the unsuccessfully decoded payload packet(s).

Inventors:	**Esswie; Ali (Calgary, CA)**
Applicant:	**Dell Products L.P.** (Round Rock, TX)
Family ID:	**1000007744944**
Appl. No.:	**18/598659**
Filed:	**March 07, 2024**

Publication Classification

Int. Cl.: **H04L1/1812** (20230101); **H04B7/185** (20060101)

U.S. Cl.:

CPC **H04L1/1819** (20130101); **H04B7/18513** (20130101);

Background/Summary

BACKGROUND

[0001] The ‘New Radio’ (NR) terminology that is associated with fifth generation mobile wireless communication systems (“5G”) refers to technical aspects used in wireless radio access networks (“RAN”) that comprise several quality of service classes (“QoS”), including ultrareliable and low latency communications (“URLLC”), enhanced mobile broadband (“eMBB”), and massive machine type communication (“mMTC”). The URLLC QoS class is associated with a stringent latency requirement (e.g., low latency or low signal/message delay) and a high reliability of radio performance, while conventional eMBB use cases may be associated with high-capacity wireless communications, which may permit less stringent latency requirements (e.g., higher latency than URLLC) and less reliable radio performance as compared to URLLC. Performance requirements for mMTC may be lower than for eMBB use cases. Some use case applications involving mobile devices or mobile user equipment such as smart phones, wireless tablets, smart watches, and the like, may impose on a given RAN resource loads, or demands, that vary. A RAN node may activate a network energy saving mode to reduce power consumption.

SUMMARY

[0002] The following presents a simplified summary of the disclosed subject matter in order to provide a basic understanding of some of the various embodiments. This summary is not an extensive overview of the various embodiments. It is intended neither to identify key or critical elements of the various embodiments nor to delineate the scope of the various embodiments. Its sole purpose is to present some concepts of the disclosure in a streamlined form as a prelude to the more detailed description that is presented later.

[0003] In an example embodiment, a method may comprise facilitating, by a terrestrial radio network node comprising at least one processor, receiving, from a user equipment, a delegated hybrid automatic repeat request comprising at least one protocol data unit identifier indicative of at least one unsuccessfully received protocol data unit corresponding to at least one non-terrestrial downlink traffic flow. The method may further comprise facilitating, by the terrestrial radio network node, transmitting, to a non-terrestrial network component, a non-terrestrial payload fetch request comprising the at least one protocol data unit identifier, facilitating, by the terrestrial radio network node, receiving, from the non-terrestrial network component, at least one retransmitted protocol data unit corresponding to the at least one protocol data unit identifier, and facilitating, by the terrestrial radio network node, transmitting, to the user equipment, the at least one retransmitted protocol data unit.

[0004] The at least one unsuccessfully received protocol data unit may be directed to the user equipment. In an embodiment, the user equipment may be a first user equipment and the at least one unsuccessfully received protocol data unit may be directed to a second user equipment. The first user equipment may be an extended reality processing unit and the second user equipment maybe an extended reality appliance with respect to which the extended reality processing unit, or the first user equipment, is facilitating relaying of traffic with a non-terrestrial network node.

[0005] The non-terrestrial network component may comprise a non-terrestrial network gateway, a user plane function, or a non-terrestrial network node. The user plane function may be a component of a core network. The at least one unsuccessfully received protocol data unit may have been transmitted to the user equipment by a non-terrestrial network node.

[0006] The at least one unsuccessfully received protocol data unit may comprise at least one packet of a group of non-terrestrial downlink packets, and wherein the at least one protocol data unit identifier is indicative of the group of non-terrestrial downlink packets. The at least one protocol data unit identifier may comprise a sequence number.

[0007] The delegated hybrid automatic repeat request may further comprise at least one of: a target non-terrestrial network node identifier corresponding to a non-terrestrial network node that transmitted the at least one unsuccessfully received protocol data unit, a redundancy version indication indicative of a redundancy version corresponding to the at least one unsuccessfully received protocol data unit, or a modulation and coding indication indicative of modulation and coding information used to transmit the at least one unsuccessfully received protocol data unit.
[0008] The modulation and coding information may correspond to a modulation and coding scheme. The facilitating of the transmitting of the at least one retransmitted protocol data unit may comprise facilitating the transmitting according to the modulation and coding information used to transmit the at least one unsuccessfully received protocol data unit.
[0009] The redundancy version may be a first redundancy version, and wherein the facilitating of the transmitting of the at least one retransmitted protocol data unit comprises facilitating the transmitting according to a second redundancy version that is sequentially subsequent to the first redundancy version. In an embodiment, the modulation and coding information may correspond to a default modulation and decoding scheme and the second redundancy version may be a next available redundancy version that is not sequentially related to the first redundancy version.
[0010] In an embodiment, the method may comprise facilitating, by the terrestrial radio network node, transmitting, to the user equipment, a delegated hybrid automatic repeat request combine indication indicative to the user equipment to enable combination of the at least one unsuccessfully received protocol data unit and the at least one retransmitted protocol data unit. The delegated hybrid automatic repeat request combine indication may be transmitted to the user equipment, by the terrestrial network node, along with, or in conjunction with, the at least one retransmitted protocol data unit.
[0011] In an embodiment, a redundancy version indication may be absent from the delegated hybrid automatic repeat request. the method may further comprise determining, by the terrestrial radio network node, at least one channel condition parameter metric corresponding to a communication link between the terrestrial radio network node and the user equipment. Based on the at least one channel condition parameter metric, the method may further comprise determining, by the terrestrial radio network node, a modulation and coding scheme, wherein the facilitating of the transmitting of the at least one retransmitted protocol data unit comprises facilitating the transmitting according to the modulation and coding scheme.
[0012] In an embodiment, the method may further comprise determining, by the terrestrial radio network node, a redundancy version to result in a locally-determined redundancy version, wherein the facilitating of the transmitting of the at least one retransmitted protocol data unit is facilitating the transmitting according to the locally-determined redundancy version. The method may further comprise facilitating, by the terrestrial radio network node, transmitting, to the user equipment, a non-combine indication indicative to the user equipment to avoid combining the at least one retransmitted protocol data unit with the at least one unsuccessfully received protocol data unit. The non-combine indication may be transmitted if the delegated hybrid automatic repeat request does not indicate modulation and coding information or redundancy version information corresponding to the at least one protocol data unit, terrestrial port node to the user equipment, that is being retransmitted by the terrestrial network node.
[0013] In another example embodiment, a terrestrial radio network node may comprise at least one processor configured to process executable instructions that, when executed by the at least one processor, facilitate performance of operations that may comprise receiving, from a user equipment, a delegated hybrid automatic repeat request comprising at least one packet identifier indicative of at least one unsuccessfully received packet corresponding to at least one non-terrestrial downlink traffic flow. The operations may further comprise transmitting, to a core network component, a non-terrestrial payload fetch request comprising the at least one packet identifier. Responsive to the non-terrestrial payload fetch request, the operations may further

comprise receiving, from the core network component, at least one retransmitted packet corresponding to the at least one packet. The operations may comprise transmitting, to the user equipment, the at least one retransmitted packet.

[0014] The delegated hybrid automatic repeat request may further comprise at least one of: a target non-terrestrial network node identifier corresponding to a non-terrestrial network node that transmitted the at least one unsuccessfully received packet, a redundancy version indication indicative of a redundancy version corresponding to the at least one unsuccessfully received packet, or a modulation and coding indication indicative of modulation and coding information used to transmit the at least one unsuccessfully received packet.

[0015] In an embodiment, the redundancy version may be a first redundancy version. The transmitting of the at least one retransmitted packet may be facilitated according to a second redundancy version that is sequentially subsequent to the first redundancy version and according to the modulation and coding information used to transmit the at least one unsuccessfully received packet.

[0016] In an embodiment, the operations may further comprise facilitating, by the terrestrial radio network node, transmitting, to the user equipment, a delegated hybrid automatic repeat request combine indication indicative to the user equipment to enable combining of the at least one unsuccessfully received packet and the at least one retransmitted packet.

[0017] In yet another example embodiment, a non-transitory machine-readable medium may comprising executable instructions that, when executed by at least one processor of a terrestrial radio network node, facilitate performance of operations that may comprise receiving, from a user equipment, a delegated hybrid automatic repeat request comprising at least one packet identifier indicative of at least one unsuccessfully received packet, corresponding to at least one non-terrestrial downlink traffic flow, transmitted to the user equipment by a non-terrestrial network node and a non-terrestrial network node identifier corresponding to the non-terrestrial network node. The operations may further comprise transmitting, to a core network component, a non-terrestrial payload fetch request comprising the at least one packet identifier and the non-terrestrial network node identifier, receiving, from the core network component, at least one retransmitted packet corresponding to the at least one packet identifier, and transmitting, to the user equipment, the at least one retransmitted packet.

[0018] In an embodiment, the delegated hybrid automatic repeat request may further comprise at least one of: a redundancy version indication indicative of a redundancy version corresponding to the at least one unsuccessfully received packet, or a modulation and coding indication indicative of modulation and coding information used to transmit the at least one unsuccessfully received packet.

[0019] In an embodiment, the redundancy version may be a first redundancy version. The transmitting of the at least one retransmitted packet may be facilitated according to a second redundancy version that is sequentially subsequent, with respect to a circular sequence, to the first redundancy version and according to the modulation and coding information used to transmit the at least one unsuccessfully received packet. The operations may further comprise transmitting, to the user equipment, a delegated hybrid automatic repeat request combine indication indicative to the user equipment to enable the at least one unsuccessfully received protocol data unit and the at least one retransmitted packet to be combined.

[0020] In an embodiment, a redundancy version indication may be absent from the delegated hybrid automatic repeat request. The operations may further comprise determining at least one channel condition parameter metric corresponding to a communication link between the terrestrial radio network node and the user equipment. Based on the at least one channel condition parameter metric, the operations may further comprise determining, by the terrestrial radio network node, a modulation and coding scheme, wherein the transmitting of the at least one retransmitted packet is facilitated according to the modulation and coding scheme, determining, by the terrestrial radio network node, a redundancy version to result in a locally-determined redundancy version. The transmitting of the at least one retransmitted packet may be further facilitated according to the locally-determined redundancy version. The operations may further comprise transmitting, to the user equipment, a non-combine indication indicative to the user equipment to prevent the at least one retransmitted packet and the at least one unsuccessfully received packet from being combined.

[0021] In another example method embodiment, a method may comprise receiving, by a user equipment comprising at least one processor from a non-terrestrial network node, a delegated hybrid automatic repeat request configuration comprising a delegated hybrid automatic repeat request indication indicative to the user equipment to enable transmission of a delegated hybrid automatic repeat request to a terrestrial radio network node with respect to unsuccessfully receiving at least one protocol data unit transmitted to the user equipment by the non-terrestrial network node. The method may further comprise determining, by the user equipment, that at least one protocol data unit, corresponding to at least one non-terrestrial downlink traffic flow and received from the non-terrestrial network node, is unsuccessfully received to result in at least one unsuccessfully received protocol data unit. The method may further comprise transmitting, by the user equipment to the terrestrial radio network node, a delegated hybrid automatic repeat request comprising at least one protocol data unit identifier indicative of the at least one unsuccessfully received protocol data unit. Responsive to the transmitting of the delegated hybrid automatic repeat request, the method may further comprise receiving, by the user equipment from the terrestrial radio network node, at least one terrestrial retransmitted protocol data unit corresponding to the at least one protocol data unit identifier. The delegated hybrid automatic repeat request configuration is received via a radio resource control signal.

[0022] The delegated hybrid automatic repeat request configuration may comprise at least one non-terrestrial downlink traffic identifier indicative of at least one downlink bearer or at least one downlink traffic flow with respect to which transmission, by the user equipment to the terrestrial radio network node, of the at least one delegated hybrid automatic repeat request is enabled.

[0023] In an embodiment, the method may further comprise determining, by the user equipment, a remaining time budget corresponding to retransmission, by the non-terrestrial network node to the user equipment, of the at least one unsuccessfully received protocol data unit to result in a determined remaining time budget, analyzing, by the user equipment, the determined remaining time budget with respect to a latency criterion to result in an analyzed determined remaining time budget, and determining, by the user equipment, to facilitate the transmitting of the delegated hybrid automatic repeat request based on the analyzed determined remaining time budget being determined to violate the latency criterion. The latency criterion may be associated with a Quality-of-Service. The latency criterion may be associated with a traffic flow with respect to which the at least one unsuccessfully received protocol data unit corresponds. The latency criterion may be a time budget within which the at least one protocol data unit, that is unsuccessfully received, is to be successfully received and decoded by the user equipment.

[0024] In an embodiment, the determining of the determined remaining time budget may be based on a roundtrip time to transmit, by the user equipment to the non-terrestrial network node, a hybrid automatic repeat request negative acknowledgement, and to transmit, by the non-terrestrial network node to the user equipment in response to a hybrid automatic repeat request negative acknowledgement, a non-terrestrial at least one retransmitted protocol data unit corresponding to the at least one protocol data unit identifier. The determining of the determined remaining time budget may be further based on a decoding time corresponding to decoding of at least one of the at least one protocol data unit corresponding to at least one non-terrestrial downlink traffic flow.

[0025] The delegated hybrid automatic repeat request further comprises at least one of: a non-terrestrial network node identifier corresponding to the non-terrestrial network node; a redundancy version indication indicative of a redundancy version corresponding to transmission by the non-terrestrial network node of the determined at least one unsuccessfully received protocol data unit; or a modulation and coding scheme indication indicative of a modulation and coding scheme corresponding to transmission by the non-terrestrial network node of the at least one unsuccessfully received protocol data unit.

[0026] In an embodiment, the redundancy version may be a first redundancy version. The at least one terrestrial retransmitted protocol data unit may be transmitted by the terrestrial radio network node to the user equipment according to a second redundancy version that is sequentially subsequent to the first redundancy version. The method may further comprise combining, by the user equipment, the at least one terrestrial retransmitted protocol data unit with the at least one unsuccessfully received protocol data unit based on the second redundancy version and the second redundancy version respectively, to result in at least one combined protocol data unit, and decoding, by the user equipment, the at least one combined protocol data unit.

[0027] The at least one protocol data unit identifier may comprise at least one sequence number corresponding to the at least one unsuccessfully received protocol data unit. The at least one protocol data unit identifier may be usable by the terrestrial radio network node to retrieve, from a core network component, the at least one terrestrial retransmitted protocol data unit.

[0028] In an embodiment, the at least one unsuccessfully received protocol data unit may be at least one first unsuccessfully received non-terrestrial protocol data unit. The determined remaining time budget may be a first determined remaining time budget. The analyzed determined remaining time budget may be a first analyzed determined remaining time budget. The, at least one terrestrial retransmitted protocol data unit may be at least one first terrestrial retransmitted protocol data unit. The at least one protocol data unit identifier may be at least one first protocol data unit identifier. The method may further comprise determining, by the user equipment, that at least one second non-terrestrial protocol data unit, received from the non-terrestrial network node, is unsuccessfully received to result in at least one second unsuccessfully received protocol data unit, determining, by the user equipment, a second remaining time budget corresponding to retransmission, by the non-terrestrial network node to the user equipment, of the at least one second unsuccessfully received protocol data unit to result in a second determined remaining time budget, and analyzing, by the user equipment, the second determined remaining time budget with respect to the latency criterion to result in a second analyzed determined remaining time budget. Based on the second analyzed determined remaining time budget being determined to satisfy the latency criterion, the method may further comprise transmitting, by the user equipment to the non-terrestrial network node, a hybrid automatic repeat request negative acknowledgement comprising at least one second protocol data unit identifier indicative of the at least one second unsuccessfully received protocol data unit. Responsive to the transmitting of the hybrid automatic repeat request negative acknowledgment, the method may further comprise receiving, by the user equipment from the non-terrestrial network node, at least one second terrestrial retransmitted protocol data unit corresponding to the at least one second protocol data unit identifier.

[0029] In an embodiment, a redundancy version indication may be absent from the delegated hybrid automatic repeat request. The receiving of the at least one terrestrial retransmitted protocol data unit corresponding to the at least one protocol data unit identifier may be facilitated according to a default redundancy version. The method may comprise flushing, by the user equipment from a memory corresponding to the user equipment, the at least one unsuccessfully received protocol data unit, and decoding, by the user equipment, the at least one terrestrial retransmitted protocol data unit.

[0030] In an embodiment, a modulation and coding scheme indication may be absent from the delegated hybrid automatic repeat request. The receiving of the at least one terrestrial retransmitted protocol data unit corresponding to the at least one protocol data unit identifier may be facilitated according to a default modulation and coding scheme configured in the user equipment.

[0031] In an embodiment, a modulation and coding scheme indication may be absent from the delegated hybrid automatic repeat request. the method may further comprise determining, by the user equipment, a modulation and coding scheme corresponding to the terrestrial radio network node based on at least one channel condition parameter metric corresponding to a communication link between the terrestrial radio network node and the user equipment to result in a determined modulation and coding scheme, wherein the receiving of the at least one terrestrial retransmitted protocol data unit corresponding to the at least one protocol data unit identifier is facilitated according to the determined modulation and coding scheme.

[0032] In another example, embodiment, a user equipment may comprise at least one processor configured to process executable instructions that, when executed by the at least one processor, facilitate performance of operations that may comprise receiving, from a non-terrestrial network node, a delegated hybrid automatic repeat request configuration comprising a delegated hybrid automatic repeat request indication indicative to the user equipment to enable transmission of a delegated hybrid automatic repeat request to a terrestrial radio network node with respect to unsuccessfully receiving at least one protocol data unit transmitted to the user equipment by the non-terrestrial network node. The operations may comprise determining that at least one packet, corresponding to at least one non-terrestrial downlink traffic flow and received from the non-terrestrial network node, is unsuccessfully received to result in at least one unsuccessfully received packet. The operations may further comprise determining a remaining time budget corresponding to retransmission, by the non-terrestrial network node to the user equipment, of the at least one unsuccessfully received packet to result in a determined remaining time budget, analyzing the determined remaining time budget with respect to a latency criterion to result in an analyzed determined remaining time budget. Based on the analyzed determined remaining time budget being determined to violate the latency criterion, the operations may further comprise transmitting, to the terrestrial radio network node, the delegated hybrid automatic repeat request comprising at least one packet identifier indicative of the at least one unsuccessfully received packet. Responsive to the transmitting of the delegated hybrid automatic repeat request, the operations may further comprise receiving, from the terrestrial radio network node, at least one terrestrial retransmitted packet corresponding to the at least one packet identifier.

[0033] In an embodiment, the at least one packet identifier may comprise at least one sequence number corresponding to the at least one unsuccessfully received packet. The at least one packet identifier is usable by the terrestrial radio network node to retrieve, from a core network component, the at least one terrestrial retransmitted packet.

[0034] In an embodiment, the at least one packet identifier may comprise at least one sequence number corresponding to the at least one unsuccessfully received packet. The at least one packet identifier is usable by the terrestrial radio network node to retrieve, via a non-terrestrial gateway corresponding to the non-terrestrial network node, the at least one terrestrial retransmitted packet.

[0035] In yet another example embodiment, a non-transitory machine-readable medium may comprise executable instructions that, when executed by at least processor of a user equipment, facilitate performance of operations that may comprise receiving, from a non-terrestrial network node, a delegated hybrid automatic repeat request configuration comprising a delegated hybrid automatic repeat request indication indicative to the user equipment to enable transmission of a delegated hybrid automatic repeat request to a terrestrial radio network node with respect to unsuccessfully receiving at least one packet transmitted to the user equipment by the non-terrestrial network node. The operations may further comprise determining that at least one packet, corresponding to at least one non-terrestrial downlink traffic flow, received from the non-terrestrial network node is unsuccessfully received to result in at least one packet determined to be unsuccessfully received. Based on a delegated hybrid automatic repeat request retransmission being indicated in the delegated hybrid automatic repeat request configuration being enabled with respect to the at least one non-terrestrial downlink traffic flow, the operations may further comprise determining to transmit, to the terrestrial radio network node, the delegated hybrid automatic repeat request comprising at least one packet identifier indicative of the at least one packet determined to be unsuccessfully received to result in a determined delegated hybrid automatic repeat request. The

operations may further comprise transmitting, to the terrestrial radio network node, the determined delegated hybrid automatic repeat request, and, responsive to the transmitting of the delegated hybrid automatic repeat request, receiving, from the terrestrial radio network node, at least one terrestrial retransmitted packet corresponding to the at least one packet identifier.
[0036] The determining to transmit, to the terrestrial radio network node, the delegated hybrid automatic repeat request may further comprise determining a remaining time budget corresponding to retransmission, by the non-terrestrial network node to the user equipment, of the at least one packet determined to be unsuccessfully received to result in a determined remaining time budget, analyzing the determined remaining time budget with respect to a latency criterion to result in an analyzed determined remaining time budget, and determining to facilitate the transmitting of the delegated hybrid automatic repeat request based on the analyzed determined remaining time budget being determined to violate the latency criterion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0037] FIG. **1** illustrates wireless communication system environment.
[0038] FIG. **2** illustrates an environment with a satellite base station and satellite that are capable of communication of traffic corresponding to a radio access network.
[0039] FIG. **3** illustrates an example environment with a terrestrial network node facilitating retransmission of non-terrestrial downlink traffic to a user equipment.
[0040] FIG. **4** illustrates an example delegated hybrid automatic repeat request configuration.
[0041] FIG. **5** illustrates an example delegated hybrid automatic repeat request retransmission request.
[0042] FIG. **6** illustrates a timing diagram of an example embodiment of a terrestrial radio network node facilitating retransmission of unsuccessfully decoded non-terrestrial traffic to a user equipment.
[0043] FIG. **7** illustrates a timing diagram of an example embodiment of a user equipment requesting that a terrestrial radio network node facilitate retransmission of unsuccessfully decoded non-terrestrial traffic to the user equipment.
[0044] FIG. **8** illustrates a flow diagram of an example embodiment method.
[0045] FIG. **9** illustrates a block diagram of an example method embodiment.
[0046] FIG. **10** illustrates a block diagram of an example terrestrial radio network node.
[0047] FIG. **11** illustrates a block diagram of an example non-transitory machine-readable medium embodiment.
[0048] FIG. **12** illustrates a block diagram of an example method embodiment.
[0049] FIG. **13** illustrates a block diagram of an example user equipment.
[0050] FIG. **14** illustrates a block diagram of an example non-transitory machine-readable medium embodiment.
[0051] FIG. **15** illustrates an example computer environment.
[0052] FIG. **16** illustrates a block diagram of an example wireless user equipment.
DETAILED DESCRIPTION OF THE DRAWINGS
[0053] As a preliminary matter, it will be readily understood by those persons skilled in the art that the present embodiments are susceptible of broad utility and application. Many methods, embodiments, and adaptations of the present application other than those herein described as well as many variations, modifications and equivalent arrangements, will be apparent from or reasonably suggested by the substance or scope of the various embodiments of the present application.
[0054] Accordingly, while the present application has been described herein in detail in relation to

various embodiments, it is to be understood that this disclosure is illustrative of one or more concepts expressed by the various example embodiments and is made merely for the purposes of providing a full and enabling disclosure. The following disclosure is not intended nor is to be construed to limit the present application or otherwise exclude any such other embodiments, adaptations, variations, modifications and equivalent arrangements, the present embodiments described herein being limited only by the claims appended hereto and the equivalents thereof.

[0055] As used in this disclosure, in some embodiments, the terms “component,” “system” and the like are intended to refer to, or comprise, a computer-related entity or an entity related to an operational apparatus with one or more specific functionalities, wherein the entity can be either hardware, a combination of hardware and software, software, or software in execution. As an example, a component can be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, computer-executable instructions, a program, and/or a computer. By way of illustration and not limitation, both an application running on a server and the server can be a component.

[0056] One or more components can reside within a process and/or thread of execution and a component can be localized on one computer and/or distributed between two or more computers. In addition, these components can execute from various computer readable media having various data structures stored thereon. The components can communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network such as the internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software application or firmware application executed by a processor, wherein the processor can be internal or external to the apparatus and executes at least a part of the software or firmware application. In yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, the electronic components can comprise a processor therein to execute software or firmware that confers at least in part the functionality of the electronic components. While various components have been illustrated as separate components, it will be appreciated that multiple components can be implemented as a single component, or a single component can be implemented as multiple components, without departing from example embodiments.

[0057] The term “facilitate” as used herein is in the context of a system, device or component “facilitating” one or more actions or operations, in respect of the nature of complex computing environments in which multiple components and/or multiple devices can be involved in some computing operations. Non-limiting examples of actions that may or may not involve multiple components and/or multiple devices comprise transmitting or receiving data, establishing a connection between devices, determining intermediate results toward obtaining a result, etc. In this regard, a computing device or component can facilitate an operation by playing any part in accomplishing the operation. When operations of a component are described herein, it is thus to be understood that where the operations are described as facilitated by the component, the operations can be optionally completed with the cooperation of one or more other computing devices or components, such as, but not limited to, sensors, antennae, audio and/or visual output devices, other devices, etc.

[0058] Further, the various embodiments can be implemented as a method, apparatus or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware, or any combination thereof to control a computer to implement the disclosed subject matter. The term “article of manufacture” as used herein is intended to encompass a computer program accessible from any computer-readable (or machine-readable) device or computer-readable (or machine-readable) storage/communications media. For example, computer readable storage media can comprise, but are not limited to, magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips), optical disks (e.g., compact disk (CD), digital versatile disk (DVD)), smart cards, and flash memory devices (e.g., card, stick, key drive). Of course, those skilled in the art will recognize many modifications can be made to this configuration without departing from the scope or spirit of the various embodiments.

[0059] Turning now to the figures, FIG. **1** illustrates an example of a wireless communication system **100** that supports blind decoding of PDCCH candidates or search spaces in accordance with aspects of the present disclosure. The wireless communication system **100** may include one or more base stations **105**, one or more UEs **115**, and core network **130**. In some examples, the wireless communication system **100** may be a Long Term Evolution (LTE) network, an LTE-Advanced (LTE-A) network, an LTE-A Pro network, or a New Radio (NR) network. In some examples, the wireless communication system **100** may support enhanced broadband communications, ultra-reliable (e.g., mission critical) communications, low latency communications, communications with low-cost and low-complexity devices, or any combination thereof. As shown in the figure, examples of UEs **115** may include smart phones, automobiles or other vehicles, or drones or other aircraft. Another example of a UE may be a virtual reality appliance **117**, such as smart glasses, a virtual reality headset, an augmented reality headset, and other similar devices that may provide images, video, audio, touch sensation, taste, or smell sensation to a wearer. A UE, such as VR appliance **117**, may transmit or receive wireless signals with a RAN base station **105** via a long-range wireless link **125**, or the UE/VR appliance may receive or transmit wireless signals via a short-range wireless link **137**, which may comprise a wireless link with a UE device **115**, such as a Bluetooth link, a Wi-Fi link, and the like. A UE, such as appliance **117**, may simultaneously communicate via multiple wireless links, such as over a link **125** with a base station **105** and over a short-range wireless link. VR appliance **117** may also communicate with a wireless UE via a cable, or other wired connection. A RAN, or a component thereof, or a gateway **106**, or a component thereof, may be implemented by one or more computer components that may be described in reference to FIG. **15**.

[0060] Continuing with discussion of FIG. **1**, base stations **105** may be dispersed throughout a geographic area to form the wireless communication system **100** and may be devices in different forms or having different capabilities. Base stations **105** and UEs **115** may wirelessly communicate via one or more communication links **125**. A base station **105** may be referred to as a RAN node. Each base station **105** may provide a coverage area **110** over which UEs **115** and the base station **105** may establish one or more communication links **125**. Coverage area **110** may be an example of a geographic area over which a base station **105** and a UE **115** may support the communication of signals according to one or more radio access technologies.

[0061] UEs **115** may be dispersed throughout a coverage area **110** of the wireless communication system **100**, and each UE **115** may be stationary, or mobile, or both at different times. UEs **115** may be devices in different forms or having different capabilities. Some example UEs **115** are illustrated in FIG. **1**. UEs **115** described herein may be able to communicate with various types of devices, such as other UEs **115**, base stations **105**, or network equipment (e.g., core network nodes, relay devices, integrated access and backhaul (IAB) nodes, or other network equipment), as shown in FIG. **1**.

[0062] Base stations **105** may communicate with the core network **130**, or with one another, or both. For example, base stations **105** may interface with core network **130** through one or more backhaul links **120** (e.g., via an S1, N2, N3, or other interface). Base stations **105** may communicate with one another over the backhaul links **120** (e.g., via an X2, Xn, or other interface) either directly (e.g., directly between base stations **105**), or indirectly (e.g., via core network **130**), or both. In some examples, backhaul links **120** may comprise one or more wireless links.

[0063] One or more of base stations **105** described herein may include or may be referred to by a person having ordinary skill in the art as a base transceiver station, a radio base station, an access point, a radio transceiver, a NodeB, an eNodeB (eNB), a next-generation NodeB or a giga-NodeB (either of which may be referred to as a bNodeB or gNB), a Home NodeB, a Home eNodeB, or other suitable terminology.

[0064] A UE **115** may include or may be referred to as a mobile device, a wireless device, a remote device, a handheld device, or a subscriber device, a wireless transmit receive unit ("WTRU"), or some other suitable terminology, where the "device" may also be referred to as a unit, a station, a terminal, or a client, among other examples. A UE **115** may also include or may be referred to as a personal electronic device such as a cellular phone, a personal digital assistant (PDA), a tablet computer, a laptop computer, a personal computer, an end extended reality appliance, an extended reality processing unit, or a router. In some examples, a UE **115** may include or be referred to as a wireless local loop (WLL) station, an Internet of Things (IoT) device, an Internet of Everything (IoE) device, or a machine type communications (MTC) device, among other examples, which may be implemented in various objects such as appliances, vehicles, or smart meters, among other examples.

[0065] UEs **115** may be able to communicate with various types of devices, such as other UEs **115** that may sometimes act as relays as well as base stations **105** and the network equipment including macro eNBs or gNBs, small cell eNBs or gNBs, or relay base stations, among other examples, as shown in FIG. **1**.

[0066] UEs **115** and base stations **105** may wirelessly communicate with one another via one or more communication links **125** over one or more carriers. The term "carrier" may refer to a set of radio frequency spectrum resources having a defined physical layer structure for supporting the communication links **125**. For example, a carrier used for a communication link **125** may include a portion of a radio frequency spectrum band (e.g., a bandwidth part (BWP)) that is operated according to one or more physical layer channels for a given radio access technology (e.g., LTE, LTE-A, LTE-A Pro, NR). Each physical layer channel may carry acquisition signaling (e.g., synchronization signals, system information), control signaling that coordinates operation for the carrier, user data, or other signaling. Wireless communication system **100** may support communication with a UE **115** using carrier aggregation or multi-carrier operation. A UE **115** may be configured with multiple downlink component carriers and one or more uplink component carriers according to a carrier aggregation configuration. Carrier aggregation may be used with both frequency division duplexing (FDD) and time division duplexing (TDD) component carriers.

[0067] In some examples (e.g., in a carrier aggregation configuration), a carrier may also have acquisition signaling or control signaling that coordinates operations for other carriers. A carrier may be associated with a frequency channel (e.g., an evolved universal mobile telecommunication system terrestrial radio access (E-UTRA) absolute radio frequency channel number (EARFCN)) and may be positioned according to a channel raster for discovery by UEs **115**. A carrier may be operated in a standalone mode where initial acquisition and connection may be conducted by UEs **115** via the carrier, or the carrier may be operated in a non-standalone mode where a connection is anchored using a different carrier (e.g., of the same or a different radio access technology).

[0068] Communication links **125** shown in wireless communication system **100** may include uplink transmissions from a UE **115** to a base station **105**, or downlink transmissions from a base station **105** to a UE **115**. Carriers may carry downlink or uplink communications (e.g., in an FDD mode) or may be configured to carry downlink and uplink communications e.g., in a TDD mode).

[0069] A carrier may be associated with a particular bandwidth of the radio frequency spectrum, and in some examples the carrier bandwidth may be referred to as a "system bandwidth" of the carrier or the wireless communication system **100**. For example, the carrier bandwidth may be one of a number of determined bandwidths for carriers of a particular radio access technology (e.g., 1.4, 3, 5, 10, 15, 20, 40, or 80 megahertz (MHz)). Devices of the wireless communication system **100** (e.g., the base stations **105**, the UEs **115**, or both) may have hardware configurations that support communications over a particular carrier bandwidth or may be configurable to support communications over one of a set of carrier bandwidths. In some examples, the wireless communication system **100** may include base stations **105** or UEs **115** that support simultaneous communications via carriers associated with multiple carrier bandwidths. In some examples, each served UE **115** may be configured for operating over portions (e.g., a sub-band, a BWP) or all of a carrier bandwidth.

[0070] Signal waveforms transmitted over a carrier may be made up of multiple subcarriers (e.g., using multi-carrier modulation (MCM) techniques such as orthogonal frequency division multiplexing (OFDM) or discrete Fourier transform spread OFDM (DFT-S-OFDM)). In a system employing MCM techniques, a resource element may consist of one symbol period (e.g., a duration of one modulation symbol) and one subcarrier, where the symbol period and subcarrier spacing are inversely related. The number of bits carried by each resource element may depend on the modulation scheme (e.g., the order of the modulation scheme, the coding rate of the modulation scheme, or both). Thus, the more resource elements that a UE **115** receives and the higher the order of the modulation scheme, the higher the data rate may be for the UE. A wireless communications resource may refer to a combination of a radio frequency spectrum resource, a time resource (e.g., a search space), or a spatial resource (e.g., spatial layers or beams), and the use of multiple spatial layers may further increase the data rate or data integrity for communications with a UE **115**.

[0071] One or more numerologies for a carrier may be supported, where a numerology may include a subcarrier spacing (Δf) and a cyclic prefix. A carrier may be divided into one or more BWPs having the same or different numerologies. In some examples, a UE **115** may be configured with multiple BWPs. In some examples, a single BWP for a carrier may be active at a given time and communications for a UE **115** may be restricted to one or more active BWPs.

[0072] The time intervals for base stations **105** or UEs **115** may be expressed in multiples of a basic time unit which may, for example, refer to a sampling period of T.sub.s=1/(Δf.sub.max.Math.N.sub.f) seconds, where Δf.sub.max may represent the maximum supported subcarrier spacing, and N.sub.f may represent the maximum supported discrete Fourier transform (DFT) size. Time intervals of a communications resource may be organized according to radio frames each having a specified duration (e.g., 10 milliseconds (ms)). Each radio frame may be identified by a system frame number (SFN) (e.g., ranging from 0 to 1023).

[0073] Each frame may include multiple consecutively numbered subframes or slots, and each subframe or slot may have the same duration. In some examples, a frame may be divided (e.g., in the time domain) into subframes, and each subframe may be further divided into a number of slots. Alternatively, each frame may include a variable number of slots, and the number of slots may depend on subcarrier spacing. Each slot may include a number of symbol periods e.g., depending on the length of the cyclic prefix prepended to each symbol period). In some wireless communication systems **100**, a slot may further be divided into multiple mini-slots containing one or more symbols. Excluding the cyclic prefix, each symbol period may contain one or more (e.g., N.sub.f) sampling periods. The duration of a symbol period may depend on the subcarrier spacing or frequency band of operation.

[0074] A subframe, a slot, a mini-slot, or a symbol may be the smallest scheduling unit (e.g., in the time domain) of the wireless communication system **100** and may be referred to as a transmission time interval (TTI). In some examples, the TTI duration (e.g., the number of symbol periods in a TTI) may be variable. Additionally, or alternatively, the smallest scheduling unit of the wireless communication system **100** may be dynamically selected (e.g., in bursts of shortened TTIs (STTIs)).

[0075] Physical channels may be multiplexed on a carrier according to various techniques. A physical control channel and a physical data channel may be multiplexed on a downlink carrier, for example, using one or more of time division multiplexing (TDM) techniques, frequency division multiplexing (FDM) techniques, or hybrid TDM-FDM techniques. A control region e.g., a control resource set (CORESET)) for a physical control channel may be defined by a number of symbol periods and may extend across the system bandwidth or a subset of the system bandwidth of the carrier. One or more control regions (e.g., CORESETs) may be configured for a set of UEs **115**. For example, one or more of UEs **115** may monitor or search control regions, or spaces, for control information according to one or more search space sets, and each search space set may include one or multiple control channel candidates in one or more aggregation levels arranged in a cascaded manner. An aggregation level for a control channel candidate may refer to a number of control channel resources (e.g., control channel elements (CCEs)) associated with encoded information for a control information format having a given payload size. Search space sets may include common search space sets configured for sending control information to multiple UEs **115** and UE-specific search space sets for sending control information to a specific UE **115**. Other search spaces and configurations for monitoring and decoding them are disclosed herein that are novel and not conventional.

[0076] A base station **105** may provide communication coverage via one or more cells, for example a macro cell, a small cell, a hot spot, or other types of cells, or any combination thereof. The term “cell” may refer to a logical communication entity used for communication with a base station **105** (e.g., over a carrier) and may be associated with an identifier for distinguishing neighboring cells (e.g., a physical cell identifier (PCID), a virtual cell identifier (VCID), or others). In some examples, a cell may also refer to a geographic coverage area **110** or a portion of a geographic coverage area **110** (e.g., a sector) over which the logical communication entity operates. Such cells may range from smaller areas (e.g., a structure, a subset of structure) to larger areas depending on various factors such as the capabilities of a base station **105**. For example, a cell may be or include a building, a subset of a building, or exterior spaces between or overlapping with geographic coverage areas **110**, among other examples.

[0077] A macro cell generally covers a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by UEs **115** with service subscriptions with the network provider supporting the macro cell. A small cell may be associated with a lower-powered base station **105**, as compared with a macro cell, and a small cell may operate in the same or different (e.g., licensed, unlicensed) frequency bands as macro cells. Small cells may provide unrestricted access to the UEs **115** with service subscriptions with the network provider or may provide restricted access to the UEs **115** having an association with the small cell (e.g., UEs **115** in a closed subscriber group (CSG), UEs **115** associated with users in a home or office). A base station **105** may support one or multiple cells and may also support communications over the one or more cells using one or more component carriers.

[0078] In some examples, a carrier may support multiple cells, and different cells may be configured according to different protocol types (e.g., MTC, narrowband IoT (NB-IoT), enhanced mobile broadband (eMBB)) that may provide access for different types of devices.

[0079] In some examples, a base station **105** may be movable and therefore provide communication coverage for a moving geographic coverage area **110**. In some examples, different geographic coverage areas **110** associated with different technologies may overlap, but the different geographic coverage areas **110** may be supported by the same base station **105**. In other examples, the overlapping geographic coverage areas **110** associated with different technologies may be supported by different base stations **105**. The wireless communication system **100** may include, for example, a heterogeneous network in which different types of the base stations **105** provide coverage for various geographic coverage areas **110** using the same or different radio access technologies.

[0080] The wireless communication system **100** may support synchronous or asynchronous operation. For synchronous operation, the base stations **105** may have similar frame timings, and transmissions from different base stations **105** may be approximately aligned in time. For asynchronous operation, base stations **105** may have different frame timings, and transmissions from different base stations **105** may, in some examples, not be aligned in time. The techniques described herein may be used for either synchronous or asynchronous operations.

[0081] Some UEs **115**, such as MTC or IoT devices, may be low cost or low complexity devices and may provide for automated communication between machines (e.g., via Machine-to-Machine (M2M) communication). M2M communication or MTC may refer to data communication technologies that allow devices to communicate with one another or a base station **105** without human intervention. In some examples, M2M communication or MTC may include communications from devices that integrate sensors or meters to measure or capture information and relay such information to a central server or application program that makes use of the information or presents the information to humans interacting with the application program. Some UEs **115** may be designed to collect information or enable automated behavior of machines or other devices. Examples of applications for MTC devices include smart metering, inventory monitoring, water level monitoring, equipment monitoring, healthcare monitoring, wildlife monitoring, weather and geological event monitoring, fleet management and tracking, remote security sensing, physical access control, and transaction-based business charging.
[0082] Some UEs **115** may be configured to employ operating modes that reduce power consumption, such as half-duplex communications (e.g., a mode that supports one-way communication via transmission or reception, but not transmission and reception simultaneously). In some examples, half-duplex communications may be performed at a reduced peak rate. Other power conservation techniques for the UEs **115** include entering a power saving deep sleep mode when not engaging in active communications, operating over a limited bandwidth (e.g., according to narrowband communications), or a combination of these techniques. For example, some UEs **115** may be configured for operation using a narrowband protocol type that is associated with a defined portion or range (e.g., set of subcarriers or resource blocks (RBs)) within a carrier, within a guard-band of a carrier, or outside of a carrier.
[0083] The wireless communication system **100** may be configured to support ultra-reliable communications or low-latency communications, or various combinations thereof. For example, the wireless communication system **100** may be configured to support ultra-reliable low-latency communications (URLLC) or mission critical communications. UEs **115** may be designed to support ultra-reliable, low-latency, or critical functions (e.g., mission critical functions). Ultra-reliable communications may include private communication or group communication and may be supported by one or more mission critical services such as mission critical push-to-talk (MCPTT), mission critical video (MCVideo), or mission critical data (MCData). Support for mission critical functions may include prioritization of services, and mission critical services may be used for public safety or general commercial applications. The terms ultra-reliable, low-latency, mission critical, and ultra-reliable low-latency may be used interchangeably herein.
[0084] In some examples, a UE **115** may also be able to communicate directly with other UEs **115** over a device-to-device (D2D) communication link **135** (e.g., using a peer-to-peer (P2P) or D2D protocol). Communication link **135** may comprise a sidelink communication link. One or more UEs **115** utilizing D2D communications may be within the geographic coverage area **110** of a base station **105**. Other UEs **115** in such a group may be outside the geographic coverage area **110** of a base station **105** or be otherwise unable to receive transmissions from a base station **105**. In some examples, groups of UEs **115** communicating via D2D communications may utilize a one-to-many (1:M) system in which a UE transmits to every other UE in the group. In some examples, a base station **105** facilitates the scheduling of resources for D2D communications. In other cases, D2D communications are carried out between UEs **115** without the involvement of a base station **105**.
[0085] In some systems, the D2D communication link **135** may be an example of a communication channel, such as a sidelink communication channel, between vehicles (e.g., UEs **115**). In some examples, vehicles may communicate using vehicle-to-everything (V2X) communications, vehicle-to-vehicle (V2V) communications, or some combination of these. A vehicle may signal information related to traffic conditions, signal scheduling, weather, safety, emergencies, or any other information relevant to a V2X system. In some examples, vehicles in a V2X system may

communicate with roadside infrastructure, such as roadside units, or with the network via one or more RAN network nodes (e.g., base stations **105**) using vehicle-to-network (V2N) communications, or with both.

[0086] The core network **130** may provide user authentication, access authorization, tracking, Internet Protocol (IP) connectivity, and other access, routing, or mobility functions. Core network **130** may be an evolved packet core (EPC) or 5G core (5GC), which may include at least one control plane entity that manages access and mobility (e.g., a mobility management entity (MME), an access and mobility management function (AMF)) and at least one user plane entity that routes packets or interconnects to external networks (e.g., a serving gateway (S-GW), a Packet Data Network (PDN) gateway (P-GW), or a user plane function (UPF)). The control plane entity may manage non-access stratum (NAS) functions such as mobility, authentication, and bearer management for UEs **115** that are served by the base stations **105** associated with core network **130**. User IP packets may be transferred through the user plane entity, which may provide IP address allocation as well as other functions. The user plane entity may be connected to IP services **150** for one or more network operators. IP services **150** may comprise access to the Internet, Intranet(s), an IP Multimedia Subsystem (IMS), or a Packet-Switched Streaming Service.

[0087] Some of the network devices, such as a base station **105**, may include subcomponents such as an access network entity **140**, which may be an example of an access node controller (ANC). Each access network entity **140** may communicate with the UEs **115** through one or more other access network transmission entities **145**, which may be referred to as radio heads, smart radio heads, or transmission/reception points (TRPs). Each access network transmission entity **145** may include one or more antenna panels. In some configurations, various functions of each access network entity **140** or base station **105** may be distributed across various network devices e.g., radio heads and ANCs) or consolidated into a single network device (e.g., a base station **105**).

[0088] The wireless communication system **100** may operate using one or more frequency bands, typically in the range of 300 megahertz (MHz) to 300 gigahertz (GHz). Generally, the region from 300 MHz to 3 GHz is known as the ultra-high frequency (UHF) region or decimeter band because the wavelengths range from approximately one decimeter to one meter in length. The UHF waves may be blocked or redirected by buildings and environmental features, but the waves may penetrate structures sufficiently for a macro cell to provide service to UEs **115** located indoors. The transmission of UHF waves may be associated with smaller antennas and shorter ranges (e.g., less than 100 kilometers) compared to transmission using the smaller frequencies and longer waves of the high frequency (HF) or very high frequency (VHF) portion of the spectrum below 300 MHZ.

[0089] The wireless communication system **100** may also operate in a super high frequency (SHF) region using frequency bands from 3 GHz to 30 GHz, also known as the centimeter band, or in an extremely high frequency (EHF) region of the spectrum (e.g., from 30 GHz to 300 GHZ), also known as the millimeter band. In some examples, the wireless communication system **100** may support millimeter wave (mmW) communications between the UEs **115** and the base stations **105**, and EHF antennas of the respective devices may be smaller and more closely spaced than UHF antennas. In some examples, this may facilitate use of antenna arrays within a device. The propagation of EHF transmissions, however, may be subject to even greater atmospheric attenuation and shorter range than SHF or UHF transmissions. The techniques disclosed herein may be employed across transmissions that use one or more different frequency regions, and designated use of bands across these frequency regions may differ by country or regulating body.

[0090] The wireless communication system **100** may utilize both licensed and unlicensed radio frequency spectrum bands. For example, the wireless communication system **100** may employ License Assisted Access (LAA), LTE-Unlicensed (LTE-U) radio access technology, or NR technology in an unlicensed band such as the 5 GHz industrial, scientific, and medical (ISM) band. When operating in unlicensed radio frequency spectrum bands, devices such as base stations **105** and UEs **115** may employ carrier sensing for collision detection and avoidance. In some examples, operations in unlicensed bands may be based on a carrier aggregation configuration in conjunction with component carriers operating in a licensed band (e.g., LAA). Operations in unlicensed spectrum may include downlink transmissions, uplink transmissions, P2P transmissions, or D2D transmissions, among other examples.

[0091] A base station **105** or a UE **115** may be equipped with multiple antennas, which may be used to employ techniques such as transmit diversity, receive diversity, multiple-input multiple-output (MIMO) communications, or beamforming. The antennas of a base station **105** or a UE **115** may be located within one or more antenna arrays or antenna panels, which may support MIMO operations or transmit or receive beamforming. For example, one or more base station antennas or antenna arrays may be co-located at an antenna assembly, such as an antenna tower. In some examples, antennas or antenna arrays associated with a base station **105** may be located in diverse geographic locations. A base station **105** may have an antenna array with a number of rows and columns of antenna ports that the base station **105** may use to support beamforming of communications with a UE **115**. Likewise, a UE **115** may have one or more antenna arrays that may support various MIMO or beamforming operations. Additionally, or alternatively, an antenna panel may support radio frequency beamforming for a signal transmitted via an antenna port.

[0092] Base stations **105** or UEs **115** may use MIMO communications to exploit multipath signal propagation and increase the spectral efficiency by transmitting or receiving multiple signals via different spatial layers. Such techniques may be referred to as spatial multiplexing. The multiple signals may, for example, be transmitted by the transmitting device via different antennas or different combinations of antennas. Likewise, the multiple signals may be received by the receiving device via different antennas or different combinations of antennas. Each of the multiple signals may be referred to as a separate spatial stream and may carry bits associated with the same data stream (e.g., the same codeword) or different data streams (e.g., different codewords). Different spatial layers may be associated with different antenna ports used for channel measurement and reporting. MIMO techniques include single-user MIMO (SU-MIMO), where multiple spatial layers are transmitted to the same receiving device, and multiple-user MIMO (MU-MIMO), where multiple spatial layers are transmitted to multiple devices.

[0093] Beamforming, which may also be referred to as spatial filtering, directional transmission, or directional reception, is a signal processing technique that may be used at a transmitting device or a receiving device (e.g., a base station **105**, a UE **115**) to shape or steer an antenna beam (e.g., a transmit beam, a receive beam) along a spatial path between the transmitting device and the receiving device. Beamforming may be achieved by combining the signals communicated via antenna elements of an antenna array such that some signals propagating at particular orientations with respect to an antenna array experience constructive interference while others experience destructive interference. The adjustment of signals communicated via the antenna elements may include a transmitting device or a receiving device applying amplitude offsets, phase offsets, or both to signals carried via the antenna elements associated with the device. The adjustments associated with each of the antenna elements may be defined by a beamforming weight set associated with a particular orientation (e.g., with respect to the antenna array of the transmitting device or receiving device, or with respect to some other orientation).

[0094] A base station **105** or a UE **115** may use beam sweeping techniques as part of beam forming operations. For example, a base station **105** may use multiple antennas or antenna arrays (e.g., antenna panels) to conduct beamforming operations for directional communications with a UE **115**. Some signals (e.g., synchronization signals, reference signals, beam selection signals, or other control signals) may be transmitted by a base station **105** multiple times in different directions. For example, a base station **105** may transmit a signal according to different beamforming weight sets associated with different directions of transmission. Transmissions in different beam directions may be used to identify (e.g., by a transmitting device, such as a base station **105**, or by a receiving device, such as a UE **115**) a beam direction for later transmission or reception by the base station

105.

[0095] Some signals, such as data signals associated with a particular receiving device, may be transmitted by a base station **105** in a single beam direction (e.g., a direction associated with the receiving device, such as a UE **115**). In some examples, the beam direction associated with transmissions along a single beam direction may be determined based on a signal that was transmitted in one or more beam directions. For example, a UE **115** may receive one or more of the signals transmitted by a base station **105** in different directions and may report to the base station an indication of the signal that the UE **115** received with a highest signal quality or an otherwise acceptable signal quality.

[0096] In some examples, transmissions by a device (e.g., by a base station **105** or a UE **115**) may be performed using multiple beam directions, and the device may use a combination of digital precoding or radio frequency beamforming to generate a combined beam for transmission (e.g., from a base station **105** to a UE **115**). A UE **115** may report feedback that indicates precoding weights for one or more beam directions, and the feedback may correspond to a configured number of beams across a system bandwidth or one or more sub-bands. A base station **105** may transmit a reference signal (e.g., a cell-specific reference signal (CRS), a channel state information reference signal (CSI-RS)), which may be precoded or unprecoded. A UE **115** may provide feedback for beam selection, which may be a precoding matrix indicator (PMI) or codebook-based feedback (e.g., a multi-panel type codebook, a linear combination type codebook, a port selection type codebook). Although these techniques are described with reference to signals transmitted in one or more directions by a base station **105**, a UE **115** may employ similar techniques for transmitting signals multiple times in different directions (e.g., for identifying a beam direction for subsequent transmission or reception by the UE **115**) or for transmitting a signal in a single direction (e.g., for transmitting data to a receiving device).

[0097] A receiving device (e.g., a UE **115**) may try multiple receive configurations (e.g., directional listening) when receiving various signals from the base station **105**, such as synchronization signals, reference signals, beam selection signals, or other control signals. For example, a receiving device may try multiple receive directions by receiving via different antenna subarrays, by processing received signals according to different antenna subarrays, by receiving according to different receive beamforming weight sets (e.g., different directional listening weight sets) applied to signals received at multiple antenna elements of an antenna array, or by processing received signals according to different receive beamforming weight sets applied to signals received at multiple antenna elements of an antenna array, any of which may be referred to as “listening” according to different receive configurations or receive directions. In some examples, a receiving device may use a single receive configuration to receive along a single beam direction e.g., when receiving a data signal). The single receive configuration may be aligned in a beam direction determined based on listening according to different receive configuration directions (e.g., a beam direction determined to have a highest signal strength, highest signal-to-noise ratio (SNR), or otherwise acceptable signal quality based on listening according to multiple beam directions).

[0098] The wireless communication system **100** may be a packet-based network that operates according to a layered protocol stack. In the user plane, communications at the bearer or Packet Data Convergence Protocol (PDCP) layer may be IP-based. A Radio Link Control (RLC) layer may perform packet segmentation and reassembly to communicate over logical channels. A Medium Access Control (MAC) layer may perform priority handling and multiplexing of logical channels into transport channels. The MAC layer may also use error detection techniques, error correction techniques, or both to support retransmissions at the MAC layer to improve link efficiency. In the control plane, the Radio Resource Control (RRC) protocol layer may provide establishment, configuration, and maintenance of an RRC connection between a UE **115** and a base station **105** or a core network **130** supporting radio bearers for user plane data. At the physical layer, transport channels may be mapped to physical channels.

[0099] The UEs **115** and the base stations **105** may support retransmissions of data to increase the likelihood that data is received successfully. Hybrid automatic repeat request (HARQ) feedback is one technique for increasing the likelihood that data is received correctly over a communication link **125**. HARQ may include a combination of error detection (e.g., using a cyclic redundancy check (CRC)), forward error correction (FEC), and retransmission (e.g., automatic repeat request (ARQ)). HARQ may improve throughput at the MAC layer in poor radio conditions (e.g., low signal-to-noise conditions). In some examples, a device may support same-slot HARQ feedback, where the device may provide HARQ feedback in a specific slot for data received in a previous symbol in the slot. In other cases, the device may provide HARQ feedback in a subsequent slot, or according to some other time interval.
[0100] The evolution of communication networks has witnessed remarkable advancements over the past decades. A significant extension of 5G's potential may lie beyond the conventional terrestrial infrastructure, giving rise to what are known as Non-Terrestrial Networks ("NTN").
[0101] Non-Terrestrial Networks may encompass a diverse range of technologies and architectures that may comprise space-based, airborne, and maritime platforms to enhance global communication capabilities. Integration of 5G and non-terrestrial environments may facilitate connectivity being established, maintained, and optimized to remote and underserved regions.
[0102] Satellites equipped with 5G capabilities constitute an aspect of 5G NTN. Satellites, positioned in low Earth orbit ("LEO"), medium Earth orbit ("MEO"), or geostationary orbit ("GEO"), may form an intricate web of interconnected nodes. The satellites can provide widespread coverage, offering high-speed data connections, low latency communication, and global mobility. Satellites may facilitate broadband access in rural and remote areas, disaster-stricken regions, and on moving vehicles, ships, and aircraft, thus bridging the digital divide.
[0103] Satellite-based NTN can bridge connectivity gaps in remote and rural areas, provide disaster recovery communication, and offer enhanced coverage for maritime and aeronautical services. High-altitude platforms and drones equipped with cellular capabilities can serve as temporary network relays for events, emergencies, or areas with signal-strength coverage deficiencies. such applications may benefit not only traditional voice and data services but also for technologies, such as, for example, Internet of Things ("IoT"), wherein connectivity is typically a desirable, or a fundamental requirement.
[0104] A non-terrestrial base station **106**, which may comprise a satellite antenna, may be coupled to core network **130**. Non-terrestrial base station **106** may communicate with satellite **107**, which may communicate with a user equipment **115**. Non-terrestrial base station **106**, which may be referred to as a non-terrestrial network gateway, and satellite **107** may facilitate delivering traffic corresponding to a radio access network, which may comprise RAN nodes **105**, core network **130**, backhaul links **120**, and long-range wireless links **125**, to user equipment that may be located beyond coverage of a RAN node **105**. Links **121** between RAN nodes **105** and satellite base station/gateway **106** may comprise coaxial, fiber, or wireless links that may be similar to links **120**. Links **122** and **124** to satellite node **107**, and links **123** from satellite/node **107** to UE **115**, may comprise line-of-sight microwave signal transmission. A UE **115** may be configured with at least one antenna, or at least one processor, to facilitate transmitting or receiving microwave signals to/from satellite node **107**. Description of herein, or reference to herein, a radio node or a radio network node may be a description or a reference to either a terrestrial RAN node **105**, a non-terrestrial gateway **106**, a non-terrestrial satellite node **107**, or a combination of one or more of a terrestrial RAN node, a non-terrestrial gateway, or a non-terrestrial satellite. A terrestrial network node may be referred to as a "TN" node. Reference to a satellite node, or a non-terrestrial network node ("NTN node"), may comprise a reference to satellite **107**, base station gateway **106**, or a combination of satellite **107** and base station/gateway **106**.
[0105] It will be appreciated that although an NTN node may benefit the most from embodiments disclosed herein, techniques disclosed herein may be of benefit to a ground-based RAN node.

Thus, use of “radio network node” may be interpreted as referring to a ground-based RAN node or to a satellite node, which may comprise a gateway **106** or a satellite **107**.

[0106] NTNs can enhance the limited coverage of ground RANs, which makes NTNs cost efficient in remote rural areas, mountainous areas, and generally where ground cellular deployments are either not possible or not cost efficient.

[0107] Incorporating RAN node functionality on board a satellite to facilitate serving user equipment may give rise to performance-related problems, such as, for example, the exceptionally large payload retransmission round trip time (“RTT”), compared to payload retransmission RTT corresponding to conventional terrestrial-based communication between a TN RAN node and a user equipment. The larger RTT associated with NTN transmission may diminish the benefit of transmission of traffic via a non-terrestrial network. When an NTN payload packet, transmitted by a non-terrestrial network node to an NTN-capable user equipment, is not successfully decoded by an NTN device, a negative acknowledgement (“NACK”) is typically transmitted by the user equipment device towards the satellite/NTN node, and, responsive to receiving the NACK, the NTN node retransmits the corresponding payload. Thus, a full NTN RTT delay would be the amount of time between the NTN-capable UE transmitting the NACK and receiving the payload retransmitted in response to receiving the NACK. A full RTT with respect to a UE transmitting a NACK to an NTN node and receiving a packet retransmitted in response thereto may be hundreds of milliseconds. For latency-critical downlink traffic flows a RTT hundreds of milliseconds in duration may result in a packet, retransmitted in response to a NACK, being a useless packet with respect to an application that may be executing at an NTN-capable user equipment, even before being delivered to the NTN-capable user equipment.

[0108] According to embodiments disclosed herein, latency-intolerant NTN downlink packet retransmissions may be dynamically routed via a TN/ground RAN node instead of the packets being delivered an NTN node, which may significantly reduce an experienced packet retransmission RTT delay due to a lower propagation delay corresponding to retransmission via the TN node compared to retransmission via the NTN node. According to embodiments disclosed herein, a ground/TN RAN node may retransmit NTN traffic payload, even if the TN RAN node did not facilitate attempted delivery of a previous transmission of payload to which the retransmitted payload corresponds. Because a previous payload transmission, facilitated by an NTN node, and respective one or more payload retransmissions, facilitated by a TN RAN node, are related, a mode of triggering a first and subsequent retransmission(s) may determine how subsequent payload versions should be combined or treated by a user equipment receiving the different retransmissions. Embodiments disclosed herein may facilitate enabling a TN RAN node to be dynamically updated with respect to basic transmission configuration information corresponding to a first transmission of payload via an NTN node such that the RAN node can fine tune triggering of the subsequent payload retransmissions to increase reception reliability corresponding to retransmitted packets.

[0109] Conventional techniques require that a RAN node retransmit packets that the node originally transmitted to a user equipment device. Embodiments disclosed herein may facilitate a TN RAN node retransmitting payload that was originally transmitted (e.g., as a first/original transmission) by another RAN node (e.g., a satellite/NTN node).

[0110] Conventional Modulation and Coding scheme (“MCS”) selection is solely based on device-specific channel conditions. Embodiments disclosed herein may enable a terrestrial radio access network node to adopt an MCS for packet retransmission that matches a first/original MCS used to transmit a packet via a non-terrestrial network node and disregard device-specific communication channel conditions corresponding to the TN RAN node in selecting an MCS. Such enablement may be beneficial wherein the original MCS level and an MCS level that might have been used for retransmission via the NTN node may be close to one another.

[0111] Turning now to FIG. **2**, the figure illustrates ground-based RAN node **105**, base station **106**, and NTN node **107**, any one or more of which may be referred to as a radio network node. In reference to some embodiments disclosed herein, reference to a TN node may comprise a reference to node **108**, which may comprise one or more of terrestrial RAN node **105** or gateway **106**. In reference to some embodiments disclosed herein, reference to an NTN node may comprise a reference to node **109**, which may comprise one or more of gateway **106** or satellite **107**. In some embodiments, a communication session with UE **115** may be served by RAN node **105**. RAN node **105** may communicate directly with satellite node **107** via communication links **124** or via gateway **106** via links **121** and **122**.

Non-Terrestrial Payload Feedback Over Terrestrial Networks and Adaptive Non-Terrestrial Packet Retransmissions.

[0112] Turning now to FIG. **3**, at act **1**, NTN-capable user equipment **115** may receive from non-terrestrial network node **107** delegated hybrid automatic repeat ("dHARQ") configuration **305**. Configuration **305** may be received via a Radio Resource Control ("RRC") signaling message. As shown in FIG. **4**, configuration **305** may comprise at least one of: a dHARQ enabled indication **415** indicative that dHARQ retransmission by a TN RAN node via a TN radio interface link is enabled; one or more associated NTN downlink traffic identifiers **420**, which may be indicative of one or more downlink bearers or one or more downlink traffic flows with respect to which dHARQ over a TN interface is enabled (as indicated by indication **415**); or at least one dHARQ mode indication **425** indicative of static or dynamic dHARQ retransmission.

[0113] Continuing with description of FIG. **3**, user equipment **115** may receive downlink traffic **310** transmitted by non-terrestrial network node **107** at act **2**. At act **3**, user equipment **115** may determine that the user equipment may have unsuccessfully received or unsuccessfully decoded at least one protocol data unit, for example a packet, corresponding to downlink traffic **310**, and may determine to request, from terrestrial radio access network node **105**, retransmission of the at least one unsuccessfully received or unsuccessfully decoded protocol data unit.

[0114] Returning to description of FIG. **4**, indications **415** and **420** may facilitate NTN RAN node **107** only enabling dHARQ operation with respect to downlink traffic that is associated with a latency-critical Quality-of-Service ("QoS") to minimize TN RAN node **105** being overwhelmed with requests for retransmission of NTN downlink payload **310** that may not be negatively impacted by retransmission by NTN node **107** and an RTT that may correspond thereto. Indication, or information element, **425** may be used to indicate to NTN device **115** to adopt dHARQ with respect to traffic indicated by indication **420**, regardless remaining real-time delay budgets, (e.g., static mode) or to dynamically determine, in real time, which NTN downlink packets or traffic flows, indicated by indication **420**, with respect to which dHARQ is activated, based on a remaining delay budget corresponding to a packet, or packets, that were unsuccessfully receive from NTN node **107** before an application, corresponding to the UE, that is expecting the unsuccessfully decoded packet is negatively impacted by failure to successfully the packet.

[0115] Continuing with discussion of act **3** shown in FIG. **3**, if NTN capable UE/WTRU device **115** determines that indication **425** in configuration **305** indicated a static dHARQ mode, and determines that at least one NTN packet, corresponding to downlink traffic **310**, has been unsuccessfully decoded, the UE may determine whether a downlink bearer or flow, with which the at least one unsuccessfully decoded packet is associated, is enabled, via indication **415** included in configuration **305**, for dHARQ retransmissions via TN RAN node **105**. If UE **115** determines that configuration **305** is indicative that dHARQ via terrestrial radio access network node **105** is enabled for the at least one unsuccessfully decoded packet, the UE may transmit a dHARQ request **315** to the terrestrial radio access network node at act **4**.

[0116] If UE **115** determines at act **3** that indication **425** in configuration **305** indicates dynamic dHARQ mode, and determines that at least one NTN packet, corresponding to downlink traffic **310**, has been unsuccessfully decoded, the UE may determine a packet remaining time budget for the retransmission of the at least one unsuccessfully decoded packet to be received before failure to successfully decode the unsuccessfully decoded packet negatively impacts an application corresponding to the UE. The remaining time budget may be determined based on an NTN HARQ RTT (e.g., RTT for UE to transmit a HARQ request to NTN node **107** and to receive a response from the NTN node to the request). A packet remaining time budget may be determined by subtracting a delay associated with decoding a first packet transmission (e.g., decoding delay associated with the at least one unsuccessfully decoded packet) and an expected NTN HARQ RTT from a maximum allowable delay budget associated with the at least one unsuccessfully decoded packet. If UE **115** determines a negative or zero packet remaining time budget, the UE may dynamically enable dHARQ with respect to the at least one unsuccessfully decoded packet (e.g., if the UE determines that requesting HARQ retransmission from the NTN node would result in violation of the delay budget associated with the at least one unsuccessfully decoded packet, the UE may request dHARQ retransmission, by TN RAN node **105**, of the at least one unsuccessfully decoded packet). If a positive remaining time budget is determined by user equipment **115** at act **3**, the user equipment may avoid transmitting a dHARQ request **315** to radio access network node **105** and may transmit a conventional HARQ request to non-terrestrial network node **107** requesting that the non-terrestrial network node retransmit the at least one unsuccessfully decoded packet.

[0117] At act **4**, on condition of unsuccessfully decoding at least one packet corresponding to a traffic flow **310** indicated by indication **420** in configuration **305** as being enabled by indication **415** for dHARQ retransmission, NTN capable WTRU/device may transmit, to TN RAN node **105**, a TN uplink control information, towards the selected and/or camped on TN RAN node, comprising NTN dHARQ retransmission request **315** requesting retransmitting the at least one unsuccessfully decoded packet. TN RAN node may receive delegated hybrid automatic repeat request **315** (shown in more detail in FIG. **5**), from NTN-capable device **115** via uplink TN radio interface **125**. Request **315** may facilitate retransmitting failed NTN downlink payload corresponding to downlink traffic **310**. As shown in FIG. **5**, request **315** may comprise information elements, such as, for example, at least one of: a target NTN/satellite ID **515**, an NTN packet sequence number or a packet data group sequence number **520** associated with a packet, or packet group, for which TN dHARQ packet retransmission is requested; a redundancy version ("RV") indication **525** indicative of a first RV applied to a first unsuccessfully decoded NTN packet; or a modulation and coding scheme ("MCS") indication **530** indicative of an MCS applied by NTN node **107** to the first unsuccessfully decoded NTN packet. In an embodiment, indications **525** and **530** may be optional, or not included in a request **315**. However, indications **525** and **530** may facilitate TN RAN node **105** fine tune transmission configuration information to be applied to NTN payload/packets requested, by request **315**, for retransmission, to match transmission configuration information applied by NTN node **107** to the first NTN payload transmission. (Conventionally, a TN RAN node is not aware of NCS or RV information corresponding to traffic transmitted by another device or equipment.) Using matching MCS and RV information to facilitate retransmission of one or more packets requested, via request **315**, for retransmission may facilitate combing of the one or more retransmitted packets with previously transmitted, and unsuccessfully decoded, versions of the packet(s).

[0118] At act **5**, TN RAN node **105** may determine NTN gateway **106** or an NTN user plane function ("UPF") entity of the core network **130**, which may facilitate delivery of NTN payload **310** with an identifier corresponding to satellite node **107** indicated in request **305**. TN RAN node **105** may transmit an NTN payload fetching request **320** via determined gateway **106** at act **6**B, or at act **6**A to core network UPF, which may be a component of core network **130**, indicative of one or more sequence numbers corresponding to the NTN packets or packet groups associated with traffic **310** requested for TN retransmission. Request **320** may be referred to as a non-terrestrial payload fetch request and may comprise at least one protocol data unit identifier corresponding to payload being requested for dHARQ retransmission. The arrowed lines indicative in FIG. **3** of request **320** being transmitted to core network **130** or gateway **106** are shown being a dashed line to indicate that request **320** may be directed to the gateway or to the core network, or both.

[0119] Unlike conventional ground-based retransmission, wherein a RAN node may be requested to retransmit a packet that the RAN node itself previously transmitted, since TN RAN node **105** shown in the FIG. **3** did not transmit the previously transmitted payload, which was instead transmitted by NTN/satellite **107**, the TN RAN node does not have payload requested for retransmission stored in a buffer and thus requests the payload from a component of core network **130** that may be facilitating downlink traffic **310**, or from NTN node **107** or associated gateway **106**. It will be appreciated that transmission of request **320** may be directed at act **6**B to NTN node **107** via gateway **106**. TN RAN node **105** may receive NTN traffic payload **325** corresponding to traffic **310** (e.g., corresponding to the at least one protocol data unit identifier included in request **320**) via a backhaul interface from NTN node **106**, or NTN gateway, at act **7**B, and/or from TN core network **130** at act **7**A. The arrowed lines indicative in FIG. **3** of payload **325** being transmitted by equipment of core network **130** or gateway **106** are shown being dashed lines to indicate that payload **325** may be received from the gateway or from the core network, or both. Responsive to receiving request **315**, terrestrial radio access node **105** may transmit to UE **115**, at act **8**, retransmission payload **325**, received from gateway **106**, NTN node **107**, or equipment corresponding to core network **130**, as retransmission payload **330**, which may comprise at least one retransmitted protocol data unit corresponding to at least one protocol data unit sequence number identified in request **315**.

[0120] In an embodiment, if redundancy version ("RV") indication **525** or modulation and coding scheme ("MCS") information indication **530** are indicative of RV information or MCS information corresponding to a first, or previous, unsuccessfully decoded packet corresponding to downlink traffic **310**, being included in request **315** transmitted by UE **115** at act **4**, TN RAN node **105** may determine at act **9** to adopt the same MCS indicated in request **315** and may select a next available circularly-rotated RV level (e.g., next available with respect to an RV value indicated in request **315** as corresponding to at least one unsuccessfully decoded packet indicated in request **315**), to be used to deliver a retransmission of payload, indicated in request **315**.for NTN payload retransmission delivery. For example, TN RAN node **105** may determine at act **9** to adopt the same MCS level used by satellite/NTN node **107** to transmit a first/previous version of a packet associated with downlink traffic **310** and may disregard applying a TN MCS that corresponds to actual channel conditions associated with TN interface link **125** with UE **115** such that packet combining by UE **115** is facilitated (e.g., the user equipment can combine a previously received and unsuccessfully decoded packet and the retransmitted version of the packet). If a different MCS is used for retransmission of a packet than was used for the first/previous transmission of the packet, the UE cannot combine the previously received packet with the retransmitted packet. It will be appreciated that combining a previously transmitted packet that was unsuccessfully decoded with a retransmitted version of the packet may be advantageous because the retransmitted version may not be completely successfully decoded either but the originally/previously transmitted packet and the retransmitted packet may together include enough information to result in successfully decoded packet information and data. At act **8**, TN RAN node **105** may transmit a retransmission payload **325**, received from gateway **106**, NTN node **107**, or equipment corresponding to core network **130**, as retransmission payload **330**. Retransmission payload **330** may be accompanied by downlink control information **335**, which may be referred to as a combine indication, to be indicative to NTN-capable user equipment device **115** that dHARQ combining is enabled, wherein TN payload retransmission **330** can be combined with a previously received and unsuccessfully decoded packet/payload indicated in request **315** since retransmission payload **330** is transmitted according to the same MCS and according to an RV based on an RV used by the NTN node to transmit the previous packet/payload that was unsuccessfully decoded. Indication **335** is shown with a dashed line connecting the indication to request **315** to indicate that indication **335** may be transmitted based on MCS or RV information being contained in request **315**. On condition of request **315** not being indicative of RV or MCS information corresponding to unsuccessfully decoded payload indicated in request **315**, TN RAN node **105** may determine at act **9** to adopt a default MCS that matches channel conditions corresponding to at least one link **125** reported to TN node **105** by UE **115**. TN RAN node **105** may select a first available RV level for encoding the NTN retransmission payload. TN RAN node may indicate, via a downlink control information **336**, which may be referred to as a non-combine indication, to UE **115** that dHARQ combining is not enabled and that the first/previous NTN payload corresponding to downlink traffic **310** that was unsuccessfully decoded is to be flushed/erased from a buffer corresponding to the user equipment. Indicating to UE to flush/erase a previously received and unsuccessfully decoded packet may be useful if an MCS used to transmit, by NTN node **107**, is different from a TN MCS determined based on channel conditions corresponding to at least one link **125** between UE **115** and TN RAN node **105**, and thus would result in a decoding failure of such retransmission. Furthermore, dHARQ retransmission **330** may be delivered to, and decoded by, UE **115** faster if an MCS determined based on actual channel conditions is used than if the MCS used for the first/unsuccessfully decoded packet requested via request **315** is used. NTN capable UE/WTRU device **115** may receive retransmitted payload **330** transmitted at act **8**. If retransmitted payload **330** is successfully received and decoded, UE **115** may combine the successfully decoded retransmitted payload **330** with the previously received and unsuccessfully decoded payload if an indication **335** accompanies retransmitted payload **330**. (Even if retransmission payload is unsuccessfully received or decoded, if request **315** comprises MCS and RV information corresponding to an original/previous transmission/retransmission, the retransmission payload may be combined with the original/previous transmission/retransmission.) If retransmitted payload **330** is unsuccessfully received and decoded and accompanied by an indication **336**, UE **115** may flush the previously received and unsuccessfully decoded payload/packet and decode the received NTN payload retransmission pay load **330**. Indications **335** and **336** are illustrated with dashed lines to indicate that either indication may not necessarily be transmitted by terrestrial network node **1052** to user equipment **115**.

[0121] Turning now to FIG. **6**, the figure illustrates a timing diagram of an example embodiment method **600**. At act **605**, NTN-capable UE **115** receives from non-terrestrial network node **107** a delegated HARQ configuration (e.g., configuration **305** shown in FIG. **3**). The delegated HARQ configuration may be indicative to UE **115** that delegated hybrid automatic repeat request retransmission via terrestrial radio access network node **105** is enabled. The delegated HARQ configuration may comprise at least one of: a binary indication indicative that dHARQ packet retransmission by TN RAN node **105** via TN radio interface link(s) **125** is enabled; at least one NTN downlink bearer identifier or downlink flow identifier indication of at least one downlink bearer or downlink traffic flow with respect to which dHARQ via the TN interface link(s) is enabled; or a dHARQ mode indication indicative a static mode or a dynamic mode to be used by UE **115** to determine whether to transmit a, dynamic}. At act **610**, terrestrial network RAN node may receive a delegated hybrid automatic repeat request ("dHARQ") retransmission request, from non-terrestrial network capable device **115** via at least one uplink TN radio interface corresponding to links **125**, requesting retransmission of at least one failed NTN downlink payload packet (e.g., requesting retransmission of at least one unsuccessfully decoded packet). The dHARQ retransmission request may comprise at least one information element comprising at least one of: a target NTN/satellite identifier associated with NTN node **107**; at least one NTN packet sequence number, or packet data group sequence number, with respect to which dHARQ retransmission via TN RAN node **105** is requested; or a redundancy version indication indicative of a redundancy version corresponding to the at least one unsuccessfully decoded/failed NTN packet; or a modulation and coding scheme corresponding to the at least one unsuccessfully decoded/failed NTN packet.

[0122] At act **615**, TN RAN node **105** may determine an NTN gateway or an NTN user plane function that handles/facilitates the NTN payload being delivered by NTN node **107** to UE **115**, with respect to which the at least one unsuccessfully decoded packet/payload corresponds. At act **620**, TN RAN node **105** may transmit an NTN payload fetch request (e.g., fetch request **320** shown in FIG. **3**) to the determined gateway and/or core network UPF, indicating the sequence number(s) indicated in the dHARQ request received at act **610**. At act **625**, TN RAN node **105** may receive requested NTN payload via backhaul interface links, or microwave interface links, from NTN node **107**, and/or from the TN core network or NTN gateway determined at act **615**. At act **630**, TN node **105** may dynamically or semi-statically schedule NTN payload, received at act **625**, for retransmission via TN radio interface links **125**.

[0123] On condition of the request received at act **615** comprising RV and MCS information corresponding to the at least one unsuccessfully decoded packet with respect to which the request received at act **615** corresponds, TN RAN node **105** may determine, at act **635**, to adopt the same MCS used by NTN node **107** to transmit the at least one unsuccessfully decoded packet with respect to which the request received at act **615** corresponds, and may select a next-available (e.g., next with respect an RV indicated in the request received at act **615**) circularly-rotated RV level/value to use for retransmission to UE **115** of NTN payload received at act **625**. At act **640**, TN RAN node may transmit, to UE **115**, NTN retransmission payload. The transmitted NTN retransmission payload may comprise, or may be accompanied by, at least one downlink control information message indicative that combining of the dHARQ-requested retransmission payload transmitted it act **640** with the at least one unsuccessfully decoded packet, with respect to which the request received at act **615** corresponds, is enabled.

[0124] On condition of RV and MCS information corresponding to the at least one unsuccessfully decoded packet indicated in, and being absent from, the request received at act **615**, TN RAN node **105** may adopt a default TN MCS that matches channel conditions corresponding to that at least one link **125** that may have been reported to TN node **105** by UE **115**. TN RAN node may select a first available RV for encoding the payload received at act **625** to be retransmitted in response to receive the dHARQ retransmission request at **615**. At act **650**, TN RAN node may transmit, to UE **115** via at least one link **125**, NTN retransmission payload received at act **625**. Retransmission payload transmitted at act **650** may comprise, or may be accompanied by, a downlink control information message to be indicative to NTN-capable UE device **115** that combining of the at least one unsuccessfully decoded packet, with respect to which the request received at act **615** requests retransmission, with the retransmission payload is not enabled. User equipment **115** may flush the at least one unsuccessfully decoded packet from a buffer corresponding to the user equipment and may decode the retransmission payload received at act **650**.

[0125] Turning now to FIG. **7**, the figure illustrates a timing diagram of an example embodiment method **700**. At act **705**, NTN-capable UE **115** receives from non-terrestrial network node **107** a delegated HARQ configuration (e.g., configuration **305** shown in FIG. **3**). Configuration **305** may be received from serving NTN RAN node **107** as part of NTN radio resource control signaling, and may comprise at least one of: a dHARQ retransmission enabled indication indicative of delegating of HARQ retransmission of at least one packet, unsuccessfully received by UE **115** from NTN node **107**, via TN RAN node **105** and TN radio interface link **125** being enabled; at least one NTN downlink bearer identifiers or at least one downlink traffic flow identifiers, indicative of at least one downlink bearer or at least one downlink traffic flow with respect to which dHARQ retransmission via TN RAN node **105** is enabled; at least one dHARQ mode indication in terms of a static or dynamic mode.

[0126] In an embodiment, on condition of a dHARQ mode indication being indicative of a configured static dHARQ mode, and NTN-capable UE/WTRU **115** unsuccessfully decoding at least one packet corresponding to a downlink traffic flow being delivered by non-terrestrial network node **107**, at act **710** the NTN-capable UE/WTRU device may determine whether a downlink bearer or downlink flow, with respect to which the at least one unsuccessfully packet corresponds, is enabled, according to the configuration received at act **705**, for dHARQ retransmissions via terrestrial radio access network node **105**.

[0127] In an embodiment, on condition of a dHARQ mode indication being indicative of a configured dynamic dHARQ mode, and NTN-capable UE/WTRU **115** unsuccessfully decoding at least one packet corresponding to a downlink traffic flow being delivered by non-terrestrial network node **107**, at act **715** NTN-capable UE/WTRU device may determine a remaining time budget for retransmission of the at least one unsuccessfully decoded packet to be received before an application, corresponding to the user equipment, is negatively impacted. The remaining time budget may be determined based on an NTN HARQ round trip time (RTT). For example, a remaining time budget may be determined as a maximum allowable delay budget for packet reception corresponding to the at least one unsuccessfully decoded packet minus a delay corresponding to the unsuccessful decoding of the at least one unsuccessfully decoded packet minus an expected NTN HARQ RTT to transmit a HARQ request to NTN node **107** and to receive a retransmission of a requested unsuccessfully decoded packet from the NTN node responsive to the HARQ request. It will be appreciated that the determination given in the example assumes a first retransmission of and at least one unsuccessfully decoded packet-if the determination at act **715** is being made with respect to a subsequent retransmission of an unsuccessfully decoded packet, additional RTT value(s) and decoding delay(s) may be subtracted. On condition of UE/WTRU **115** determining at act **715** a negative and/or zero remaining time budget, the UE/WTRU may enable, at act **720**, dHARQ retransmission, by TN RAN node **105**, of the at least one unsuccessfully decoded packet. On condition of UE/WTRU **115** determining at act **715** a positive remaining time budget, the UE/WTRU may request, from NTN node **107**, HARQ retransmission of the unsuccessfully decoded packet.

[0128] If UE/WTRU has determined that at least one packet corresponding to a downlink flow being delivered by node **107** has been unsuccessfully decoded and that dHARQ retransmission of the at least one unsuccessfully decoded packet by TN RAN node **105** has been determined, at act **725** NTN-capable UE/WTRU device may transmit a TN uplink control information, toward selected and/or camped-on TN RAN node **105**, an NTN dHARQ retransmission request (e.g., request **315** described in reference to FIG. **3**), for retransmitting the at least one unsuccessfully decoded packet. The NTN dHARQ request may comprise at least one of: a target NTN/satellite identifier corresponding to NTN node **107** that delivered the at least one unsuccessfully decoded packet; at least one packet sequence number, or at least one packet data group sequence number, corresponding to the at least one unsuccessfully decoded packet for which dHARQ retransmission via TN RAN node **105** is requested; a redundancy version indication indicative a redundancy version corresponding to transmission, by NTN node **107**, of the at least one unsuccessfully decoded packet for which dHARQ retransmission is requested; or a modulation and coding scheme corresponding to transmission, by the NTN node, of the at least one unsuccessfully decoded packet for which dHARQ retransmission is requested.

[0129] At act **730**, NTN capable UE/WTRU device **115** may receive, from TN RAN node **105**, NTN payload retransmission. UE/WTRU **115** may combine and decode received NTN retransmission payload with original/previous retransmission NTN payload if combining is indicated by TN RAN as being enabled, or the UE/WTRU may flush the at least one unsuccessfully decoded packet/payload and decode the received NTN payload retransmission if combining is indicated by TN RAN node as not being enabled.

[0130] Turning now to FIG. **8**, the figure illustrates a flow diagram of an example embodiment **800**. Method **800** begins at act **805**. At act **810**, a non-terrestrial network node may transmit to a user equipment a dHARQ request configuration, for example configuration **305** described in reference to FIG. **3**. At act **815**, the user equipment may receive downlink traffic corresponding to an ongoing established communication session with the non-terrestrial network node. At act **820**, the user equipment may determine that at least one downlink packet corresponding to the ongoing communication session is unsuccessfully decoded by the user equipment. At act **825**, the user equipment may determine a remaining time budget corresponding to the unsuccessfully decoded packet, or packets, determined at act **820**. The remaining time budget may be based on a round trip time to transmit a HARQ request to the non-terrestrial network node requesting retransmission of the unsuccessfully decoded packet(s). At act **830**, the user equipment made determine whether the remaining time budget is a positive time value, or is zero or a negative time value. If a determination made at act **830** is that the remaining time budget corresponding to the unsuccessfully decoded packet(s) is a positive time value, method **800** advances to act **835**. At act **835**, the user equipment may transmit a HARQ request to the non-terrestrial network node requesting retransmission of the unsuccessfully decoded packet(s), and the user equipment may continue receiving, from the non-terrestrial network node, downlink traffic corresponding to the ongoing communication session with the non-terrestrial network node at act **815**.

[0131] Returning to description of act **830**, if the user equipment determines at act **830** that a remaining time budget corresponding to packet(s) determined to be unsuccessfully decoded at act **820** is zero or less than zero, method **800** advances to act **840**. At act **840**, the user equipment may determine, based on information contained in the request configuration received today at **810**, whether dHARQ retransmission, by a terrestrial radio access network node with respect to the packet(s) determined at act **820** to have been unsuccessfully decoded, is enabled. If a determination is made at act **840** that dHARQ retransmission by a terrestrial radio access network node is not enabled with respect to the packet(s) determined at act **820** to have been unsuccessfully decoded, method **800** advances to act **835** and the user equipment may transmit a HARQ request to the non-terrestrial network node requesting retransmission of the unsuccessfully decoded packet(s) before continuing to receive downlink traffic corresponding to the ongoing communication session with the non-terrestrial network node at act **815**.

[0132] Returning to description of act **840**, if the user equipment determines that dHARQ retransmission by a terrestrial radio access network node is enabled with respect to the packet(s) determined at act **820** to have been unsuccessfully decoded, method **800** advances to act **845**. At act **845**, the user equipment may transmit to the terrestrial radio access network node a dHARQ retransmission request. Responsive to the dHARQ retransmission request transmitted by the user equipment at act **845**, at act **850** the terrestrial radio access network node may request, from computer equipment of a core network, from a non-terrestrial gateway associated with the non-terrestrial network node, or from the non-terrestrial network node, the payload packet(s) determined at act **820** to have been unsuccessfully decoded. Responsive to requesting the unsuccessfully decoded packet(s) at **850**, at **855** the terrestrial radio access network node may receive the unsuccessfully decoded packet(s). At act **860**, the terrestrial radio access network node may determine whether the dHARQ retransmission request, transmitted by the user equipment at act **845**, comprises modulation and coding scheme information and redundancy version information corresponding to a modulation encoding scheme and a redundancy version used by the non-terrestrial network node to transmit the unsuccessfully decoded packet(s) to the user equipment. If a determination is made at act **860** that the dHARQ retransmission request transmitted by the user equipment at act **845** does not comprise modulation and coding scheme information and redundancy version information corresponding to the transmission of the packet(s) that were determined at act **820** to be unsuccessfully decoded, method **800** advances to act **865**. At act **865**, the terrestrial radio access network node may transmit, to the user equipment, according to a default modulation and coding scheme and a next available redundancy version, the payload received at act **855**. The default modulation and coding scheme may be based on channel conditions corresponding to a wireless communication link between the terrestrial radio access network node and the user equipment, wherein the channel conditions may be determined by the terrestrial radio access network node based on a radio parameter measurement report transmitted by the user equipment to the terrestrial radio access network node indicative of the channel conditions. The next available redundancy version may be a next available redundancy version to be used by the terrestrial radio access network node for any downlink transmission to the user equipment, not just to be used for retransmission of the unsuccessfully decoded packet(s). At act **870**, the user equipment may flush the unsuccessfully decoded packet(s) from a buffer corresponding to the user equipment and may decode the payload packet(s) transmitted by the terrestrial radio access network node at act **865**. Method **800** may advance from act **870** to act **885** and end.

[0133] Returning to description of act **860**, if the terrestrial radio access network node determines that dHARQ retransmission request transmitted at act **845** comprises modulation and coding scheme information and redundancy version information corresponding to transmission by the non-terrestrial radio access network node of the unsuccessfully received packet(s), method **800** may advance to act **875**. At act **875**, the terrestrial radio access network node may transmit, to the user equipment, the unsuccessfully decoded packet(s), received by the terrestrial radio access network node at act **855**, according to, or based on, modulation and coding scheme information included in the dHARQ retransmission request transmitted by the user equipment at act **845** and according to a redundancy version that is a next redundancy version with respect to the redundancy version indicated in the dHARQ retransmission request transmitted by the user equipment at act **845**. At act **880**, the user equipment may receive and decode the retransmitted payload/packet(s), transmitted at act **875**, according to the modulation and coding scheme indicated by the user equipment in the dHARQ retransmission request if the terrestrial radio access network node indicates that the user equipment is to attempt combining the unsuccessfully decoded payload/packet(s) and the retransmitted payload/packet(s) transmitted at act **875**. Method **800** advances from act **880** to act **885** and ends.

[0134] Turning now to FIG. **9**, the figure illustrates an example embodiment method **900** comprising at block **905** facilitating, by a terrestrial radio network node comprising at least one processor, receiving, from a user equipment, a delegated hybrid automatic repeat request comprising at least one protocol data unit identifier indicative of at least one unsuccessfully received protocol data unit corresponding to at least one non-terrestrial downlink traffic flow; at block **910** facilitating, by the terrestrial radio network node, transmitting, to a non-terrestrial network component, a non-terrestrial payload fetch request comprising the at least one protocol data unit identifier; at block **915** facilitating, by the terrestrial radio network node, receiving, from the non-terrestrial network component, at least one retransmitted protocol data unit corresponding to the at least one protocol data unit identifier; and at block **920** facilitating, by the terrestrial radio network node, transmitting, to the user equipment, the at least one retransmitted protocol data unit.

[0135] Turning now to FIG. **10**, the figure illustrates an example terrestrial radio network node **1000**, comprising at block **1005** at least one processor configured to process executable instructions that, when executed by the at least one processor, facilitate performance of operations, comprising receiving, from a user equipment, a delegated hybrid automatic repeat request comprising at least one packet identifier indicative of at least one unsuccessfully received packet corresponding to at least one non-terrestrial downlink traffic flow; at block **1010** transmitting, to a core network component, a non-terrestrial payload fetch request comprising the at least one packet identifier; at block **1015** responsive to the non-terrestrial payload fetch request, receiving, from the core network component, at least one retransmitted packet corresponding to the at least one packet; and at block **1020** transmitting, to the user equipment, the at least one retransmitted packet.

[0136] Turning now to FIG. **11**, the figure illustrates a non-transitory machine-readable medium **1100** comprising at block **1105** executable instructions that, when executed by at least one processor of a terrestrial radio network node, facilitate performance of operations, comprising receiving, from a user equipment, a delegated hybrid automatic repeat request comprising at least one packet identifier indicative of at least one unsuccessfully received packet, corresponding to at least one non-terrestrial downlink traffic flow, transmitted to the user equipment by a non-terrestrial network node and a non-terrestrial network node identifier corresponding to the non-terrestrial network node; at block **1110** transmitting, to a core network component, a non-terrestrial payload fetch request comprising the at least one packet identifier and the non-terrestrial network node identifier; at block **1115** receiving, from the core network component, at least one retransmitted packet corresponding to the at least one packet identifier; and at block **1120** transmitting, to the user equipment, the at least one retransmitted packet.

[0137] Turning now to FIG. **12**, the figure illustrates an example embodiment method **1200** comprising at block **1205** receiving, by a user comprising at least one processor, a delegated hybrid automatic repeat request configuration comprising a delegated hybrid automatic repeat request indication indicative to the user equipment to enable transmission of a delegated hybrid automatic repeat request to a terrestrial radio network node with respect to unsuccessfully receiving at least one protocol data unit transmitted to the user equipment by the non-terrestrial network node; at block **1210** determining, by the user equipment, that at least one protocol data unit, corresponding to at least one non-terrestrial downlink traffic flow and received from the non-terrestrial network node, is unsuccessfully received to result in at least one unsuccessfully received protocol data unit; at block **1215** transmitting, by the user equipment to the terrestrial radio network node, the delegated hybrid automatic repeat request comprising at least one protocol data unit identifier indicative of the at least one unsuccessfully received protocol data unit; and at block **1220** responsive to the transmitting of the delegated hybrid automatic repeat request, receiving, by the user equipment from the terrestrial radio network node, at least one terrestrial retransmitted protocol data unit corresponding to the at least one protocol data unit identifier.

[0138] Turning now to FIG. **13**, the figure illustrates an example user equipment **1300**, comprising at block **1305** at least one processor configured to process executable instructions that, when executed by the at least one processor, facilitate performance of operations, comprising receiving, from a non-terrestrial network node, a delegated hybrid automatic repeat request configuration comprising a delegated hybrid automatic repeat request indication indicative to the user equipment to enable transmission of a delegated hybrid automatic repeat request to a terrestrial radio network node with respect to unsuccessfully receiving at least one protocol data unit transmitted to the user equipment by the non-terrestrial network node; at block **1310** determining that at least one packet, corresponding to at least one non-terrestrial downlink traffic flow and received from the non-terrestrial network node, is unsuccessfully received to result in at least one unsuccessfully received packet; at block **1315** determining a remaining time budget corresponding to retransmission, by the non-terrestrial network node to the user equipment, of the at least one unsuccessfully received packet to result in a determined remaining time budget; at block **1320** analyzing the determined remaining time budget with respect to a latency criterion to result in an analyzed determined remaining time budget; at block **1325** based on the analyzed determined remaining time budget being determined to violate the latency criterion, transmitting, to the terrestrial radio network node, the delegated hybrid automatic repeat request comprising at least one packet identifier indicative of the at least one unsuccessfully received packet; and at block **1330** responsive to the transmitting of the delegated hybrid automatic repeat request, receiving, from the terrestrial radio network node, at least one terrestrial retransmitted packet corresponding to the at least one packet identifier.

[0139] Turning now to FIG. **14**, the figure illustrates a non-transitory machine-readable medium **1400** comprising at block **1405** executable instructions that, when executed by at least one processor of a user equipment, facilitate performance of operations, comprising receiving, from a non-terrestrial network node, a delegated hybrid automatic repeat request configuration comprising a delegated hybrid automatic repeat request indication indicative to the user equipment to enable transmission of a delegated hybrid automatic repeat request to a terrestrial radio network node with respect to unsuccessfully receiving at least one packet transmitted to the user equipment by the non-terrestrial network node; at block **1410** determining that at least one packet, corresponding to at least one non-terrestrial downlink traffic flow, received from the non-terrestrial network node is unsuccessfully received to result in at least one packet determined to be unsuccessfully received; at block **1415** based on a delegated hybrid automatic repeat request retransmission being indicated in the delegated hybrid automatic repeat request configuration being enabled with respect to the at least one non-terrestrial downlink traffic flow, determining to transmit, to the terrestrial radio network node, the delegated hybrid automatic repeat request comprising at least one packet identifier indicative of the at least one packet determined to be unsuccessfully received to result in a determined delegated hybrid automatic repeat request; at block **1420** transmitting, to the terrestrial radio network node, the determined delegated hybrid automatic repeat request; and at block **1425** responsive to the transmitting of the delegated hybrid automatic repeat request, receiving, from the terrestrial radio network node, at least one terrestrial retransmitted packet corresponding to the at least one packet identifier.

[0140] In order to provide additional context for various embodiments described herein, FIG. **15** and the following discussion are intended to provide a brief, general description of a suitable computing environment **1500** in which various embodiments of the embodiment described herein can be implemented. While embodiments have been described above in the general context of computer-executable instructions that can run on one or more computers, those skilled in the art will recognize that the embodiments can be also implemented in combination with other program modules and/or as a combination of hardware and software.

[0141] Generally, program modules include routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods can be practiced with other computer system configurations, including single-processor or multiprocessor computer systems, minicomputers, mainframe computers, IoT devices, distributed computing systems, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

[0142] The embodiments illustrated herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0143] Computing devices typically include a variety of media, which can include computer-readable storage media, machine-readable storage media, and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media or machine-readable storage media can be any available storage media that can be accessed by the computer and includes both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media or machine-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable or machine-readable instructions, program modules, structured data or unstructured data.

[0144] Computer-readable storage media can include, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD), Blu-ray disc (BD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, solid state drives or other solid state storage devices, or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per sc.

[0145] Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

[0146] Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and includes any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media include wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.
[0147] With reference again to FIG. **15**, the example environment **1500** for implementing various embodiments described herein includes a computer **1502**, the computer **1502** including a processing unit **1504**, a system memory **1506** and a system bus **1508**. The system bus **1508** couples system components including, but not limited to, the system memory **1506** to the processing unit **1504**. The processing unit **1504** can be any of various commercially available processors and may include a cache memory. Dual microprocessors and other multi-processor architectures can also be employed as the processing unit **1504**.
[0148] The system bus **1508** can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. The system memory **1506** includes ROM **1510** and RAM **1512**. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer **1502**, such as during startup. The RAM **1512** can also include a high-speed RAM such as static RAM for caching data.
[0149] Computer **1502** further includes an internal hard disk drive (HDD) **1514** (e.g., EIDE, SATA), one or more external storage devices **1516** (e.g., a magnetic floppy disk drive (FDD), a memory stick or flash drive reader, a memory card reader, etc.) and an optical disk drive **1520** (e.g., which can read or write from disk **1522**, for example a CD-ROM disc, a DVD, a BD, etc.). While the internal HDD **1514** is illustrated as located within the computer **1502**, the internal HDD **1514** can also be configured for external use in a suitable chassis (not shown). Additionally, while not shown in environment **1500**, a solid-state drive (SSD) could be used in addition to, or in place of, an HDD **1514**. The HDD **1514**, external storage device(s) **1516** and optical disk drive **1520** can be connected to the system bus **1508** by an HDD interface **1524**, an external storage interface **1526** and an optical drive interface **1528**, respectively. The interface **1524** for external drive implementations can include at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 1394 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.
[0150] The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer **1502**, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to respective types of storage devices, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, whether presently existing or developed in the future, could also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.
[0151] A number of program modules can be stored in the drives and RAM **1512**, including an operating system **1530**, one or more application programs **1532**, other program modules **1534** and program data **1536**. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM **1512**. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.
[0152] Computer **1502** can optionally comprise emulation technologies. For example, a hypervisor

(not shown) or other intermediary can emulate a hardware environment for operating system **1530**, and the emulated hardware can optionally be different from the hardware illustrated in FIG. **15**. In such an embodiment, operating system **1530** can comprise one virtual machine (VM) of multiple VMs hosted at computer **1502**. Furthermore, operating system **1530** can provide runtime environments, such as the Java runtime environment or the .NET framework, for applications **1532**. Runtime environments are consistent execution environments that allow applications **1532** to run on any operating system that includes the runtime environment. Similarly, operating system **1530** can support containers, and applications **1532** can be in the form of containers, which are lightweight, standalone, executable packages of software that include, e.g., code, runtime, system tools, system libraries and settings for an application.

[0153] Further, computer **1502** can comprise a security module, such as a trusted processing module (TPM). For instance, with a TPM, boot components hash next in time boot components, and wait for a match of results to secured values, before loading a next boot component. This process can take place at any layer in the code execution stack of computer **1502**, e.g., applied at the application execution level or at the operating system (OS) kernel level, thereby enabling security at any level of code execution.

[0154] A user can enter commands and information into the computer **1502** through one or more wired/wireless input devices, e.g., a keyboard **1538**, a touch screen **1540**, and a pointing device, such as a mouse **1542**. Other input devices (not shown) can include a microphone, an infrared (IR) remote control, a radio frequency (RF) remote control, or other remote control, a joystick, a virtual reality controller and/or virtual reality headset, a game pad, a stylus pen, an image input device, e.g., camera(s), a gesture sensor input device, a vision movement sensor input device, an emotion or facial detection device, a biometric input device, e.g., fingerprint or iris scanner, or the like. These and other input devices are often connected to the processing unit **1504** through an input device interface **1544** that can be coupled to the system bus **1508**, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a USB port, an IR interface, a BLUETOOTH® interface, etc.

[0155] A monitor **1546** or other type of display device can be also connected to the system bus **1508** via an interface, such as a video adapter **1548**. In addition to the monitor **1546**, a computer typically includes other peripheral output devices (not shown), such as speakers, printers, etc.

[0156] The computer **1502** can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) **1550**. The remote computer(s) **1550** can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically includes many or all of the elements described relative to the computer **1502**, although, for purposes of brevity, only a memory/storage device **1552** is illustrated. The logical connections depicted include wired/wireless connectivity to a local area network (LAN) **1554** and/or larger networks, e.g., a wide area network (WAN) **1556**. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the internet.

[0157] When used in a LAN networking environment, the computer **1502** can be connected to the local network **1554** through a wired and/or wireless communication network interface or adapter **1558**. The adapter **1558** can facilitate wired or wireless communication to the LAN **1554**, which can also include a wireless access point (AP) disposed thereon for communicating with the adapter **1558** in a wireless mode.

[0158] When used in a WAN networking environment, the computer **1502** can include a modem **1560** or can be connected to a communications server on the WAN **1556** via other means for establishing communications over the WAN **1556**, such as by way of the internet. The modem **1560**, which can be internal or external and a wired or wireless device, can be connected to the system bus **1508** via the input device interface **1544**. In a networked environment, program modules depicted relative to the computer **1502** or portions thereof, can be stored in the remote memory/storage device **1552**. It will be appreciated that the network connections shown are examples and other means of establishing a communications link between the computers can be used.

[0159] When used in either a LAN or WAN networking environment, the computer **1502** can access cloud storage systems or other network-based storage systems in addition to, or in place of, external storage devices **1516** as described above. Generally, a connection between the computer **1502** and a cloud storage system can be established over a LAN **1554** or WAN **1556** e.g., by the adapter **1558** or modem **1560**, respectively. Upon connecting the computer **1502** to an associated cloud storage system, the external storage interface **1526** can, with the aid of the adapter **1558** and/or modem **1560**, manage storage provided by the cloud storage system as it would other types of external storage. For instance, the external storage interface **1526** can be configured to provide access to cloud storage sources as if those sources were physically connected to the computer **1502**.

[0160] The computer **1502** can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, store shelf, etc.), and telephone. This can include Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

[0161] Turning now to FIG. **16**, the figure illustrates a block diagram of an example UE **1660**. UE **1660** may comprise a smart phone, a wireless tablet, a laptop computer with wireless capability, a wearable device, a machine device that may facilitate vehicle telematics, an intermediate XR processing unit, and the like. UE **1660** may comprise a first processor **1630**, a second processor **1632**, and a shared memory **1634**. UE **1660** may include radio front end circuitry **1662**, which may be referred to herein as a transceiver, but is understood to typically include transceiver circuitry, separate filters, and separate antennas for facilitating transmission and receiving of signals over a wireless link, such as one or more wireless links **125**, **135**, or **137** shown in FIG. **1**. Furthermore, transceiver **1662** may comprise multiple sets of circuitry or may be tunable to accommodate different frequency ranges, different modulations schemes, or different communication protocols, to facilitate long-range wireless links such as links **125**, device-to-device links, such as links **135**, and short-range wireless links, such as links **137**.

[0162] Continuing with description of FIG. **16**, UE **1660** may also include a SIM **1664**, or a SIM profile, which may comprise information stored in a memory (memory **1634** or a separate memory portion), for facilitating wireless communication with RAN **105** or core network **130** shown in FIG. **1**. FIG. **16** shows SIM **1664** as a single component in the shape of a conventional SIM card, but it will be appreciated that SIM **1664** may represent multiple SIM cards, multiple SIM profiles, or multiple eSIMs, some or all of which may be implemented in hardware or software. It will be appreciated that a SIM profile may comprise information such as security credentials (e.g., encryption keys, values that may be used to generate encryption keys, or shared values that are shared between SIM **1664** and another device, which may be a component of RAN **105**, node **107**, or core network **130** shown in FIG. **1**). A SIM profile **1664** may also comprise identifying information that is unique to the SIM, or SIM profile, such as, for example, an International Mobile Subscriber Identity (“IMSI”) or information that may make up an IMSI.

[0163] SIM **1664** is shown coupled to both first processor portion **1630** and second processor portion **1632**. Such an implementation may provide an advantage that first processor portion **1630** may not need to request or receive information or data from SIM **1664** that second processor **1632** may request, thus eliminating the use of the first processor acting as a ‘go-between’ when the second processor uses information from the SIM in performing its functions and in executing

applications. First processor **1630**, which may be a modem processor or baseband processor, is shown smaller than processor second **1632**, which may be a more sophisticated application processor than the first processor, to visually indicate the relative levels of sophistication (i.e., processing capability and performance) and corresponding relative levels of operating power consumption levels between the two processor portions. Keeping the second processor portion **1632** asleep/inactive/in a low power state when UE **1660** does not need the second processor for executing applications and processing data related to an application provides an advantage of reducing power consumption when the UE only needs to use the first processor portion **1630** while in listening mode for monitoring routine configured bearer management and mobility management/maintenance procedures, or for monitoring search spaces that the UE has been configured to monitor while the second processor portion remains inactive/asleep.
[0164] UE **1660** may also include sensors **1666**, such as, for example, temperature sensors, accelerometers, gyroscopes, barometers, moisture sensors, light sensors, and the like that may provide signals to the first processor **1630** or second processor **1632**. Output devices **1668** may comprise, for example, one or more visual displays (e.g., computer monitors, VR appliances, and the like), acoustic transducers, such as speakers or microphones, vibration components, and the like. Output devices **1668** may comprise software that interfaces with output devices, for example, visual displays, speakers, microphones, touch sensation devices, smell or taste devices, and the like, that are external to UE **1660**.
[0165] The following glossary of terms given in Table 1 may apply to one or more descriptions of embodiments disclosed herein.
TABLE-US-00001 TABLE 1 Term Definition UE User equipment WTRU Wireless transmit receive unit RAN Radio access network QoS Quality of service DRX Discontinuous reception EPI Early paging indication DCI Downlink control information SSB Synchronization signal block RS Reference signal PDCCH Physical downlink control channel PDSCH Physical downlink shared channel MUSIM Multi-SIM UE SIB System information block MIB Master information block eMBB Enhanced mobile broadband URLLC Ultra reliable and low latency communications mMTC Massive machine type communications XR Anything-reality VR Virtual reality AR Augmented reality MR Mixed reality DCI Downlink control information DMRS Demodulation reference signals QPSK Quadrature Phase Shift Keying WUS Wake up signal HARQ Hybrid automatic repeat request RRC Radio resource control C-RNTI Connected mode radio network temporary identifier CRC Cyclic redundancy check MIMO Multi input multi output UE User equipment CBR Channel busy ratio SCI Sidelink control information SBFD Sub-band full duplex CLI Cross link interference TDD Time division duplexing FDD Frequency division duplexing BS Base-station RS Reference signal CSI-RS Channel state information reference signal PTRS Phase tracking reference signal DMRS Demodulation reference signal gNB General NodeB PUCCH Physical uplink control channel PUSCH Physical uplink shared channel SRS Sounding reference signal NES Network energy saving QCI Quality class indication RSRP Reference signal received power PCI Primary cell ID BWP Bandwidth Part
[0166] The above description includes non-limiting examples of the various embodiments. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the disclosed subject matter, and one skilled in the art may recognize that further combinations and permutations of the various embodiments are possible. The disclosed subject matter is intended to embrace all such alterations, modifications, and variations that fall within the spirit and scope of the appended claims.
[0167] With regard to the various functions performed by the above-described components, devices, circuits, systems, etc., the terms (including a reference to a “means”) used to describe such components are intended to also include, unless otherwise indicated, any structure(s) which performs the specified function of the described component (e.g., a functional equivalent), even if not structurally equivalent to the disclosed structure. In addition, while a particular feature of the

disclosed subject matter may have been disclosed with respect to only one of several implementations, such feature may be combined with one or more other features of the other implementations as may be desired and advantageous for any given or particular application.

[0168] The terms “exemplary” and/or “demonstrative” or variations thereof as may be used herein are intended to mean serving as an example, instance, or illustration. For the avoidance of doubt, the subject matter disclosed herein is not limited by such examples. In addition, any aspect or design described herein as “exemplary” and/or “demonstrative” is not necessarily to be construed as preferred or advantageous over other aspects or designs, nor is it meant to preclude equivalent structures and techniques known to one skilled in the art. Furthermore, to the extent that the terms “includes,” “has,” “contains,” and other similar words are used in either the detailed description or the claims, such terms are intended to be inclusive—in a manner similar to the term “comprising” as an open transition word-without precluding any additional or other elements.

[0169] The term “or” as used herein is intended to mean an inclusive “or” rather than an exclusive “or.” For example, the phrase “A or B” is intended to include instances of A, B, and both A and B. Additionally, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless either otherwise specified or clear from the context to be directed to a singular form.

[0170] The term “set” as employed herein excludes the empty set, i.e., the set with no elements therein. Thus, a “set” in the subject disclosure includes one or more elements or entities. Likewise, the term “group” as utilized herein refers to a collection of one or more entities.

[0171] The terms “first,” “second,” “third,” and so forth, as used in the claims, unless otherwise clear by context, is for clarity only and doesn't otherwise indicate or imply any order in time. For instance, “a first determination,” “a second determination,” and “a third determination,” does not indicate or imply that the first determination is to be made before the second determination, or vice versa, etc.

[0172] The description of illustrated embodiments of the subject disclosure as provided herein, including what is described in the Abstract, is not intended to be exhaustive or to limit the disclosed embodiments to the precise forms disclosed. While specific embodiments and examples are described herein for illustrative purposes, various modifications are possible that are considered within the scope of such embodiments and examples, as one skilled in the art can recognize. In this regard, while the subject matter has been described herein in connection with various embodiments and corresponding drawings, where applicable, it is to be understood that other similar embodiments can be used or modifications and additions can be made to the described embodiments for performing the same, similar, alternative, or substitute function of the disclosed subject matter without deviating therefrom. Therefore, the disclosed subject matter should not be limited to any single embodiment described herein, but rather should be construed in breadth and scope in accordance with the appended claims below.

Claims

1. A method, comprising: facilitating, by a terrestrial radio network node comprising at least one processor, receiving, from a user equipment, a delegated hybrid automatic repeat request comprising at least one protocol data unit identifier indicative of at least one unsuccessfully received protocol data unit corresponding to at least one non-terrestrial downlink traffic flow; facilitating, by the terrestrial radio network node, transmitting, to a non-terrestrial network component, a non-terrestrial payload fetch request comprising the at least one protocol data unit identifier; facilitating, by the terrestrial radio network node, receiving, from the non-terrestrial network component, at least one retransmitted protocol data unit corresponding to the at least one protocol data unit identifier; and facilitating, by the terrestrial radio network node, transmitting, to the user equipment, the at least one retransmitted protocol data unit.

2. The method of claim 1, wherein the at least one unsuccessfully received protocol data unit is directed to the user equipment.
3. The method of claim 1, wherein the non-terrestrial network component comprises a non-terrestrial network gateway or a user plane function.
4. The method of claim 3, wherein the user plane function is a component of a core network.
5. The method of claim 1, wherein the at least one unsuccessfully received protocol data unit was transmitted to the user equipment by a non-terrestrial network node.
6. The method of claim 1, wherein the at least one unsuccessfully received protocol data unit comprises at least one packet of a group of non-terrestrial downlink packets, and wherein the at least one protocol data unit identifier is indicative of the group of non-terrestrial downlink packets.
7. The method of claim 1, wherein the delegated hybrid automatic repeat request further comprises at least one of: a target non-terrestrial network node identifier corresponding to a non-terrestrial network node that transmitted the at least one unsuccessfully received protocol data unit, a redundancy version indication indicative of a redundancy version corresponding to the at least one unsuccessfully received protocol data unit, or a modulation and coding indication indicative of modulation and coding information used to transmit the at least one unsuccessfully received protocol data unit.
8. The method of claim 7, wherein the facilitating of the transmitting of the at least one retransmitted protocol data unit comprises facilitating the transmitting according to the modulation and coding information used to transmit the at least one unsuccessfully received protocol data unit.
9. The method of claim 7, wherein the redundancy version is a first redundancy version, and wherein the facilitating of the transmitting of the at least one retransmitted protocol data unit comprises facilitating the transmitting according to a second redundancy version that is sequentially subsequent to the first redundancy version.
10. The method of claim 9, further comprising: facilitating, by the terrestrial radio network node, transmitting, to the user equipment, a delegated hybrid automatic repeat request combine indication indicative to the user equipment to enable combination of the at least one unsuccessfully received protocol data unit and the at least one retransmitted protocol data unit.
11. The method of claim 1, wherein a redundancy version indication is absent from the delegated hybrid automatic repeat request, and wherein the method further comprises: determining, by the terrestrial radio network node, at least one channel condition parameter metric corresponding to a communication link between the terrestrial radio network node and the user equipment; and based on the at least one channel condition parameter metric, determining, by the terrestrial radio network node, a modulation and coding scheme, wherein the facilitating of the transmitting of the at least one retransmitted protocol data unit comprises facilitating the transmitting according to the modulation and coding scheme.
12. The method of claim 11, further comprising: determining, by the terrestrial radio network node, a redundancy version to result in a locally-determined redundancy version, wherein the facilitating of the transmitting of the at least one retransmitted protocol data unit is facilitating the transmitting according to the locally-determined redundancy version, and wherein the method further comprises: facilitating, by the terrestrial radio network node, transmitting, to the user equipment, a non-combine indication indicative to the user equipment to avoid combining the at least one retransmitted protocol data unit with the at least one unsuccessfully received protocol data unit.
13. A terrestrial radio network node, comprising at least one processor configured to process executable instructions that, when executed by the at least one processor, facilitate performance of operations, comprising: receiving, from a user equipment, a delegated hybrid automatic repeat request comprising at least one packet identifier indicative of at least one unsuccessfully received packet corresponding to at least one non-terrestrial downlink traffic flow; transmitting, to a core network component, a non-terrestrial payload fetch request comprising the at least one packet identifier; responsive to the non-terrestrial payload fetch request, receiving, from the core network

component, at least one retransmitted packet corresponding to the at least one packet; and transmitting, to the user equipment, the at least one retransmitted packet.

14. The terrestrial radio network node of claim 13, wherein the delegated hybrid automatic repeat request further comprises at least one of: a target non-terrestrial network node identifier corresponding to a non-terrestrial network node that transmitted the at least one unsuccessfully received packet, a redundancy version indication indicative of a redundancy version corresponding to the at least one unsuccessfully received packet, or a modulation and coding indication indicative of modulation and coding information used to transmit the at least one unsuccessfully received packet.

15. The terrestrial radio network node of claim 14, wherein the redundancy version is a first redundancy version, and wherein the transmitting of the at least one retransmitted packet is facilitated according to a second redundancy version that is sequentially subsequent to the first redundancy version and according to the modulation and coding information used to transmit the at least one unsuccessfully received packet.

16. The terrestrial radio network node of claim 13, wherein the operations further comprise: facilitating, by the terrestrial radio network node, transmitting, to the user equipment, a delegated hybrid automatic repeat request combine indication indicative to the user equipment to enable combining of the at least one unsuccessfully received packet and the at least one retransmitted packet.

17. A non-transitory machine-readable medium, comprising executable instructions that, when executed by at least one processor of a terrestrial radio network node, facilitate performance of operations, comprising: receiving, from a user equipment, a delegated hybrid automatic repeat request comprising at least one packet identifier indicative of at least one unsuccessfully received packet, corresponding to at least one non-terrestrial downlink traffic flow, transmitted to the user equipment by a non-terrestrial network node and a non-terrestrial network node identifier corresponding to the non-terrestrial network node; transmitting, to a core network component, a non-terrestrial payload fetch request comprising the at least one packet identifier and the non-terrestrial network node identifier; receiving, from the core network component, at least one retransmitted packet corresponding to the at least one packet identifier; and transmitting, to the user equipment, the at least one retransmitted packet.

18. The non-transitory machine-readable medium of claim 17, wherein the delegated hybrid automatic repeat request further comprises at least one of: a redundancy version indication indicative of a redundancy version corresponding to the at least one unsuccessfully received packet, or a modulation and coding indication indicative of modulation and coding information used to transmit the at least one unsuccessfully received packet.

19. The non-transitory machine-readable medium of claim 18, wherein the redundancy version is a first redundancy version, wherein the transmitting of the at least one retransmitted packet is facilitated according to a second redundancy version that is sequentially subsequent, with respect to a circular sequence, to the first redundancy version and according to the modulation and coding information used to transmit the at least one unsuccessfully received packet, and wherein the operations further comprise: transmitting, to the user equipment, a delegated hybrid automatic repeat request combine indication indicative to the user equipment to enable the at least one unsuccessfully received protocol data unit and the at least one retransmitted packet to be combined.

20. The non-transitory machine-readable medium of claim 17, wherein a redundancy version indication is absent from the delegated hybrid automatic repeat request, and wherein the operations further comprise: determining at least one channel condition parameter metric corresponding to a communication link between the terrestrial radio network node and the user equipment; based on the at least one channel condition parameter metric, determining, by the terrestrial radio network node, a modulation and coding scheme, wherein the transmitting of the at least one retransmitted packet is facilitated according to the modulation and coding scheme; determining, by the terrestrial radio network node, a redundancy version to result in a locally-determined redundancy version, wherein the transmitting of the at least one retransmitted packet is further facilitated according to the locally-determined redundancy version; and transmitting, to the user equipment, a non-combine indication indicative to the user equipment to prevent the at least one retransmitted packet and the at least one unsuccessfully received packet from being combined.